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Scientific background document

AI Applications in Bioinformatics for Pathogen Detection via Metagenomic Sequencing

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Executive Summary

The World Health Organisation (WHO) defines the field of **health emergency and disaster preparedness as the knowledge and capacities to effectively anticipate, respond and recover from the impacts of likely, imminent or current hazardous events or conditions**. Government bodies, response and recovery organisations, communities, and individuals can carry out preparedness actions to build the capacities needed to manage all types of emergencies and achieve an orderly transition from response to sustained recovery. The **European Union is paving the way for a strong European Health Union, including among its key priorities reinforcing its resilience to cross-border health threats**. To this end, in the aftermath of the COVID-19 pandemic, the European Commission launched a new European Health Emergency Preparedness and Response Authority (HERA) for health emergencies to develop, produce, and procure medical countermeasures before and during a health crisis.

To address health emergency preparedness and response at the European Union (EU) level, **a key function of HERA is to gather health intelligence**. This health intelligence may come from threat detection activities (i.e., data collected systematically) or targeted data-gathering activities around specific health-threatening events. **A crucial element of intelligence for health emergencies and preparedness is to detect and identify (re)emerging pathogens accurately**. The study of genetic material recovered directly from bulk samples, known as **metagenomic analysis, holds great potential for enhancing pathogen identification. This facilitates outbreak detection, monitoring the spread of infection, and, eventually, improving pandemic preparedness and response.** The rapid progress in bioinformatics in recent years opened new opportunities for the early detection and surveillance of outbreaks, enabling a faster response in the crucial early stages of an outbreak and more effective ongoing monitoring of diseases.

This scientific background document explores the potential of **Artificial Intelligence (AI) in advancing metagenomic sequencing for the early detection of emerging pathogens in clinical settings and environmental studies.** The study identifies key bottlenecks in the implementation of metagenomic analysis and maps existing AI-based solutions that can potentially enhance the accuracy, speed, and interpretability of metagenomic analysis, ultimately strengthening health emergency preparedness and response.

Metagenomic sequencing has proven invaluable in identifying novel pathogens in complex samples. However, **the widespread adoption of metagenomic sequencing faces challenges** due to the complexity and volume of data generated, the poor availability of reference genome databases and the need for increased computational resources. **AI offers promising solutions to many of these bottlenecks**. AI-based tools can improve the accuracy and speed of various stages in the metagenomic workflow, from data preparation and aggregation to annotation, analysis, and visualisation. Machine learning models, including Support Vector Machines (SVMs), improve base-calling accuracy and optimise data processing. On the other hand, deep learning techniques are effective for gene prediction, functional annotation, and taxonomic classification, especially for novel or uncharacterised pathogens.

This study identifies the critical areas of the metagenomic workflow where AI can significantly enhance metagenomic analysis for emerging pathogen detection and identification:

- **Data preparation and aggregation**: AI tools can improve base-calling accuracy, particularly for certain long-read sequencing technologies like Nanopore, and optimise data processing, reducing the reliance on human experts and enhancing the overall accuracy of the workflow.
- **Annotation and analysis**: AI techniques, especially deep learning, can improve gene prediction, functional annotation, and taxonomic classification, even for novel

sequences with limited similarity to the sequences of known organisms. This is crucial for understanding the functions of unknown microorganisms and their potential impact on health. Overall, this phase has been identified as the most promising for applying AI tools.

- **Data summary and visualisation**: AI-driven tools can facilitate the visualisation of complex data clusters and enhance the interpretation of results, making metagenomic analysis more accessible to a wider range of users, including non-specialists. Unlike the two earlier phases in the workflow – Data preparation and aggregation, Annotation and analysis – this study found that traditional methods mostly suffice in data visualisation. Therefore, the need for and development of AI tools in this phase is low.

Despite the promising advancements offered by AI, the literature reviewed and the consultations with key experts highlighted **some bottlenecks that persist in the field of metagenomics which are related to data availability and accuracy, limited reference genome databases, data storage and transfer issues, and the requirement for upskilling and capacity building among professionals**.

To address these challenges, the study proposes several recommendations:

- **Facilitate data sharing and accessibility**: establishing secure data-sharing platforms and federated networks to facilitate collaboration across Member States and research institutions, ensuring access to large datasets while protecting data privacy.
    - **Standardisation of protocols**: developing standardised protocols and workflows for sample preparation, data processing, and analysis to ensure consistency and comparability across studies and across Member States.
    - **Regulatory clarity**: advocating for clear and consistent regulations implemented across Member States regarding personal data and consent for metagenomic analysis.
    - **Data licensing:** establishing a standardised licensing system for sharing genomic data to promote collaboration, secondary use, and accelerate research.
- **Continue the investment in AI research and development:** continued investment in AI-based tools and infrastructure at the EU level.
    - **Support tool validation and evaluation**: Validate and evaluate existing AI tools to build user trust and facilitate easier implementation.
    - **Support environmental metagenomics**: Increase support for developing AI solutions tailored to environmental metagenomics for better outbreak detection and prevention.
- **Enhance collaboration and promote sustainable capacity building**: Provide spaces and resources for collaboration among researchers in academia, public health laboratories, and industry to assess needs and develop solutions faster. Initiate programs to build capacity among researchers and public health professionals.

Overall, this report finds that **adopting AI-based tools in metagenomics can enhance the EU's crisis preparedness and response capabilities by enabling rapid and accurate pathogen identification**. These tools can facilitate the early detection of emerging pathogens, ensuring a timely intervention and effective resource allocation during public health crises. Moreover, **integrating AI techniques into existing surveillance and monitoring systems can strengthen the EU's ability to anticipate and mitigate the impact of infectious disease outbreaks, safeguarding public health and protecting against future health threats**.

By addressing the existing challenges and investing in AI research and development, the European Union can harness the full potential of metagenomics and AI to strengthen its health emergency preparedness and response capabilities. This will ultimately contribute to a safer and healthier future for all European citizens.

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Acronyms and abbreviations

AI	Artificial Intelligence
ATHINA	Advanced Technology for Health INtelligence and Action IT system
CBRN	Chemical, biological, radiological and nuclear
DG CONNECT	Directorate-General for Communications Networks, Content and Technology
EU	European Union
EHDS	European Health Data Space
HERA	Health Emergency Preparedness and Response Authority
HPC	High performance computing
LLM	Large Language Model
ML	Machine Learning
MAG	Metagenome-Assembled Genomes sequencing
MGS	Metagenomic sequencing
NLP	Natural Language Processing
SVMs	Support Vector Machines
WHO	World Health Organisation

Glossary

Term	Description
Amplicon Sequence Variants (ASV)	The exact sequences that were read in targeted amplicon sequencing. These sequences can differ by just one nucleotide and provide increased resolution and sensitivity compared to OTUs.
Annotation	Genome annotation is the process of deriving the structural and functional information of a protein or gene from a raw data set using different analysis, comparison, estimation, precision, and other mining techniques
Artificial intelligence (AI) system	Machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.
Assembly	The process of taking a large number of short DNA sequences and putting them back together to create a representation of the original chromosomes from which the DNA originated.
Base-calling	Base calling is the process of algorithmically deciding the incorporated nucleotide from the signal intensities that are detected during sequencing process.
Binning / Read binning	A process used to classify DNA sequences obtained from metagenomic sequencing into discrete groups, or bins, based on their similarity to each other
Contig	A contig is a set of DNA segments or sequences that overlap in a way that provides a contiguous representation of a genomic region.
Deep Learning	Subset of ML that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.
Genotype	An individual's collection of genes. Genotypes are expressed when its genetic code is used to make protein and RNA molecules. This expression contributes to the individual's observable traits, called the phenotype.
Greedy algorithm	A greedy algorithm is an algorithmic strategy that makes the best optimal choice at each small stage with the goal of this eventually leading to a globally optimum solution. This means that the algorithm picks the best solution at the moment without regard for

	consequences. It picks the best immediate output, but does not consider the big picture, hence it is considered greedy.
High performance computing (HPC)	Technology that uses clusters of powerful processors that work in parallel to process massive multi-dimensional data sets, also known as big data, and solve complex problems at extremely high speeds.
Machine Learning (ML)	An approach to artificial intelligence in which a computer algorithm (a set of rules and procedures) is developed to analyse and make predictions from data that is fed into the system.
Metagenome assembled genomes (MAG)	A species-level microbial genome constructed from community-level metagenomic data from a microbiome sample.
Metagenomics	The study of genetic material recovered directly from bulk samples, analysing the collective genomes of microbial communities to understand their diversity, function, and evolution.
Metagenomic next generation sequencing	Next generation sequencing is any of several high-throughput sequencing methods whereby billions of nucleic acid fragments can be simultaneously and independently sequenced.
Next-generation sequencing (NGS)	A high-throughput sequencing method that enables billions of nucleic acid fragments to be simultaneously and independently sequenced
Operational Taxonomic Unit (OTU)	A group of closely related individuals which are arranged together based on the similarity of specific sequences (usually the 16S rRNA gene).
Pathogen	A bacterium, virus, or other microorganism that can cause disease.
Phenotype	An individual's observable traits, such as eye colour, height, and blood type. Some phenotypic traits are largely determined by environmental factors, but others are largely determined by the individual's genes.
Reads	In next-generation sequencing, a read refers to the DNA sequence from one fragment (a small section of DNA).
Sequencing	Sequencing DNA means determining the order of the four chemical building blocks - called "bases" - that make up the DNA molecule.
Sequencing - Targeted amplicon sequencing	A targeted sequencing approach focusing on specific regions of DNA. It is based on PCR-amplification of variable regions of DNA within conserved phylogenetic or functional marker genes.
Sequencing - Whole genome sequencing (WGS)	A powerful method that allows researchers to analyse the complete genetic sequence of an individual, including both coding and non-coding regions, in order to gain insights into the genetic basis of complex traits and diseases.
Supervised learning	Machine Learning technique characterised by its use of labelled data sets to train algorithms to classify data or predict outcomes accurately.
Support Vector Machines (SVM)	A support vector machine (SVM) is a supervised machine learning algorithm that classifies data by finding an optimal line or hyperplane that maximizes the distance between each class in an N-dimensional space. The SVM algorithm is widely used in machine learning as it can handle both linear and nonlinear classification tasks.
Taxonomy	The science of classifying organisms into hierarchical groups based on shared characteristics, often involving identifying and categorizing microorganisms based on their genetic sequences.
Unsupervised learning	Machine Learning technique characterised by its use of unlabelled data sets to train algorithms to discover hidden patterns or data groupings without the need for human intervention.

1. Introduction

1.1 Study scope

The EU4Health programme 2021-2027 (European Commission, 2021) is paving the way for a strong European Health Union, including among its key priorities reinforcing the European Union's resilience to cross-border health threats. Among the top priorities for achieving a strong European Health Union are protection, prevention, preparedness and response against human health hazards. To this end, in the aftermath of the COVID-19 pandemic, the European Commission launched a new European Health Emergency Preparedness and Response Authority (HERA) for health emergencies to develop, produce, and procure medical countermeasures before and during a health crisis (European Commission, 2024). As outlined in the HERA Work Plan 2024, **HERA is advancing its strategy to promote advanced research and development of medical countermeasures and related technologies aimed at addressing priority cross-border health threats**. This includes tackling challenges like shortages of medical countermeasures and chemical, biological, radiological, and nuclear (CBRN) threats. The strategy also aims to address market challenges and failures, bolster EU strategic autonomy and strengthen knowledge, skills and HERA's international role in relation to medical countermeasures (European Commission, 2024)**.**

To deal with health emergency preparedness and response at the European Union (EU) level, **a key function of HERA is to gather health intelligence,** defined as the process of moving from data through knowledge synthesis to action with the specific aim of early detection for effective response (WHO, 2019). Health intelligence may come from monitoring activities (data collected systematically) or targeted data-gathering activities around specific health-threatening events. A **crucial element of intelligence for health emergencies and preparedness is the accurate identification of emerging or new pathogens**. Early identification of emerging or new pathogens through solid monitoring systems or targeted testing enables a timely response to health threats. This approach facilitates the rapid implementation of specific interventions to prevent the spread of these threats and supports the selection of effective treatment and prevention measures. Pathogens are often subject to various selection pressures that drive constant changes in their characteristics. Therefore, in addition to detecting well-characterised pathogens, it is essential to accurately identify new and evolving pathogens in complex samples.

In this regard, metagenomic sequencing – the study of genetic material recovered directly from bulk samples – allows the analysis of the collective genomes of microbial communities to understand their diversity, functionality, and evolution (Wheeler, et al., 2023)**. Metagenomic analysis holds great potential for pandemic preparedness and novel pathogen identification because it enables rapid discovery of known and unknown pathogens in a complex sample without prior knowledge of the composition of the sample and facilitates outbreak detection and the monitoring of the spread of infections**. The rapid progress in bioinformatics in recent years opened new opportunities for the early detection and surveillance of outbreaks, enabling a faster response in the crucial early stages of an outbreak and more effective ongoing monitoring of diseases.

In parallel to HERA's work, **the European Commission is leading a coordinated effort to embrace Artificial Intelligence solutions that aim to address public health and global crises.** Initiatives like the Repository of Artificial Intelligence (AI) and Robotics solutions or the establishment of a Testing and Experimentation Facility (TEF) dedicated to the health sector (European Commission, 2024) were launched to disseminate knowledge and accelerate the adoption of cutting-edge technologies (Fontes, Corrigan, & Lütge, 2023). The availability and widespread access to increasing volumes of data has fostered the development and use of AI-based solutions in the health sector. Due to the large amounts of data that must be processed in a metagenomics study, **metagenomics could greatly benefit from further bioinformatic**

innovations, including novel AI-based approaches, which could play an important role in supporting HERA's mission to improve EU health security.

At the EU level, **HERA is currently supporting innovations in the hardware aspects of clinical metagenomics**, enabling the development of devices that can be used directly at the point of care. In fact, HERA is contributing to this global effort by cooperating with the U. S. Biomedical Advanced Research and Development Authority (BARDA) to collectively increase financial support for advancements in metagenomic next-generation sequencing - a high-throughput sequencing method that enables billions of nucleic acid fragments to be simultaneously and independently sequenced - toward agnostic diagnostics (n.d., 2023).

1.2 Study objectives

This report aims to contribute to the current knowledge by **mapping the software tools, with a special focus on AI, that can be useful in pathogen identification and health emergency and preparedness.** The report will investigate how further innovation in the software aspects could improve the analysis of metagenomic data, making it more accurate, faster and more easily interpretable. The report will also identify specific bottlenecks in this field where further innovation is needed, such as the wealth of data generated by metagenomic sequencing and its storage requirements. Additionally, it will explore whether AI-based tools can help reduce genomic data storage and transfer requirements. In identifying the areas where further innovation is needed, the report will also take into consideration the activities and initiatives at the EU level that may affect this area of innovation, such as the developments within the ATHINA project and the implementation of the AI Act or the European Health Data Space (EHDS).

This scientific background document aims to contribute to HERA's implementation of the 2024 Work Plan and its preparedness activities.

More specifically, this scientific study has the following three main objectives:

1. Assess the **potential** of AI in metagenomics and identify how AI-based tools can make metagenomic sequencing more accurate, faster, and easily interpretable.
2. Identify the specific **bottlenecks** in the field of metagenomic data analysis where further innovation is needed.
3. Explore how **novel AI-based tools** can help make systematic metagenomic surveillance possible.

1.3 Structure of the report

The current section provides an overview of the state of the art, especially how metagenomics is currently used in health preparedness and the role played by AI-based tools in the early detection of pathogens. An overview of the initial bottlenecks identified at the preliminary stage of the study is provided at the end of this section.

Following the overview of the state of the art, Section 2 presents the results of the literature review, which aimed to establish a concrete understanding of the current state of the field, including the identified bottlenecks. Additionally, the literature review sought to inform the direction of consultation activities by assessing the gaps between the research questions and the existing literature. The findings are organised according to the three main steps of the metagenomic workflow:

- AI tools for the preparation and aggregation of genomic sequences in metagenomic studies;

- AI tools for the annotation and analysis of genomic sequences;
- AI tools for metagenomics data summarisation and visualisation.

The Results section is complemented by the main outcomes of consultation activities with key experts in the field. The consultation activities also gathered additional input that was not directly correlated to the research questions, i.e. the priorities and funding opportunities of the EC, the collaboration between industry and researchers, and some additional considerations on the use of AI, which are presented in Subsection 2.3.

Section 3 of the report is dedicated to discussions and conclusions derived from a triangulation of the key findings from the literature review and consultation activities. It presents the major implications of these findings and suggests directions for future research and policy actions.

Finally, the report includes one Annex, which provides a detailed overview of the methodology applied for the literature review and for the consultation activities.

1.4 Overview of the state of the art

The World Health Organisation (WHO) defines the field of **health emergency and disaster preparedness as the knowledge and capacities to effectively anticipate, respond and recover from the impacts of likely, imminent or current hazardous events or conditions** (WHO, Health Emergencies, 2024). Governments, response and recovery organisations, communities, and individuals can carry out preparedness actions to build the capacities needed to manage all types of emergencies and achieve an orderly transition from response to sustained recovery. Based on a sound analysis of disaster risks and good linkages with early warning systems, these actions include contingency planning, stockpiling of equipment and supplies, establishment and testing of coordination mechanisms, risk awareness raising and public education of protective behaviours, and associated training and field exercises.

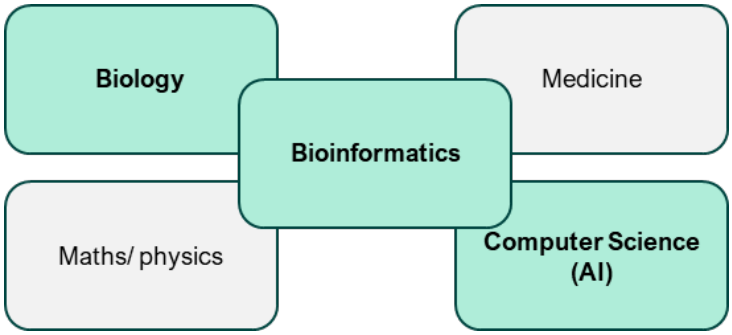
Approximately 70-80% of high-risk pathogens originate in animals and then spread between animals and humans (WHO, n.d.). Some of these are known, but many are unknown and are currently circulating among animal species around the world. Thus far, metagenomics has been proven useful in detecting novel species and strains, outbreak transmission patterns, and complicated microbiome diseases (i.e. diseases that are conventionally untestable), where other approaches have failed (Govender, 2021). **The role of metagenomics in pandemic preparedness is to serve as the early pathogen monitoring system to detect outbreaks of unknown aetiology and to enable responses as early as possible** (Govender, 2021). The importance of metagenomics in this field is also proven by the significant scientific progress that has been made to standardise approaches with guidelines, notable examples being the STROBE-metagenomics guidelines and the Strengthening The Organisation and Reporting of Microbiome Studies (STORMS) checklist (von Elm, 2008; Mirzayi, et al., 2021).

Metagenomic sequencing is revolutionising the way we identify and manage infectious disease outbreaks. This technology is capable of detecting pathogens in clinical samples that might have been missed by conventional tests, leading to more accurate diagnoses and treatments. Additionally, portable sequencing devices have enabled rapid metagenomic analysis in the field during outbreaks of Usutu, Ebola, Zika, Lassa fever, and Yellow fever, leading to more effective interventions (Govender, 2021). Moreover, metagenomics has a leading role in the analysis of complex samples from environmental surveillance programmes for emerging pathogen detection (Piombo, et al., 2021). **By analysing the genetic make-up of pathogens, metagenomics can trace transmission routes with high precision, thus being extremely helpful both at the levels of individual patients and the public health system**.

The Human Genome Project (NIH, 2024) in the early 2000s marked the beginning of a significant increase in biological data generation, prompting since numerous sequencing projects on other organisms and advancing research in genomics.

The corresponding huge demand for analysis and interpretation of these data has been managed by the evolving science of bioinformatics. Bioinformatics is defined as the application of tools of computation and analysis to capture and interpret biological data (National Human Genome Research Institute, 2024) (Figure 1). It lies in an interdisciplinary field which harnesses computer science, mathematics, physics, and biology (Bayat, 2002). Between 1995 and 2001, two groundbreaking achievements jumpstarted the growth of bioinformatics as a global venture. In 1995, a novel strategy for random sequencing of the whole genome, the so-called "shotgun" technique, allowed the mapping of the full genome of *Haemophilus influenzae* (Fleischmann, 1995). Six years later, the entire human genome was mapped for the first time (International Human Genome Sequencing Consortium, 2001). **This wealth of sequencing data generated demanded bioinformatics tools to be created and utilised, and computer networks have been created to allow easy access to biological data and enable the development of software programs for faster analysis.**

***Figure 1**. Interplay between bioinformatics, biology and computer science (AI)*



***Source:** authors' elaboration based on Bayat, A. (2002)*

Bioinformatics still continues to be a global venture, as proved by the launch of the EU-U.S. Health Task on May 2023 about priorities in the area of cancer, global health threats, and strengthening the global health architecture, thus including an active collaboration between HERA and the US-based Biomedical Advanced Research and Development Authority (BARDA) on clinical metagenomic next-generation sequencing diagnostics for rapid, pathogen-agnostic identification of infectious threats.

1.4.1 Use of metagenomics in health preparedness

Health preparedness, as defined by the WHO and other preeminent experts in the field (RAND Corporation, 2007), involves strategies and actions that are taken as precautionary measures in the face of potential disasters to protect human health and lives. Worldwide disasters could manifest in the form of pandemics caused by the spread of (re)emerging pathogens. Thus, early detection and surveillance for re-emerging pathogens is the mainstay of initiating infection control measures.

The classic approach for pathogen identification uses culture-dependent methods where microorganisms are cultured and isolated before the genomic analysis (Batool & Galloway-Peña, 2023). However, for novel unknown pathogens, culture-independent methods – such as metagenomics, broad-range PCR with traditional sequencing, and shotgun proteomics – have shown great promise. These methods analyse the genomic material of the entire population of microorganisms in a sample. Among these, metagenomics sequencing offers a comprehensive and unbiased method to detect and analyse the genetic material (DNA or RNA) of microorganisms, including bacteria, viruses, fungi, and protists, without the need for culture-based methods (Han, et al., 2019). Therefore, **metagenomic sequencing can be a valuable tool in this regard, thus being an integral part of pandemic preparedness plans, thanks to its help in disease surveillance, early detection and identification of novel pathogens**

(Aarestrup, Bonten, & Koopmans, 2021). Specifically, pathogen detection and identification are key elements in outbreak control of human, animal, and plant diseases (WHO).

Metagenomic sequencing for the early detection of pathogens can be performed in a variety of scenarios, including its use in clinical diagnostics, environmental surveillance campaigns, and a third, novel, scenario in human pathogens detection (non-clinical). The origin of the sample (i.e., clinical settings, environment) is important, as it defines both the type of methodological approach but also the expected output that metagenomic analysis can deliver. In **clinical settings**, metagenomic analysis helps identify known pathogens by comparing samples to reference genome databases. This approach is particularly useful for detecting pathogens that are rarely tested for or missed by routine methods due to unusual disease associations or primer mismatches (Carbo, et al., 2020). The samples analysed in clinical metagenomics are complex, with a high proportion of human genetic information. The role of metagenomics in this context is to provide fast and accurate identification of pathogens to inform on the right decision for treatment and containment measures to healthcare providers. Beyond clinical samples, metagenomic sequencing is indispensable in **environmental surveillance**, enabling the identification of novel pathogens found in water, soil or biological material samples, often not catalogued yet in accessible reference genome databases. This is crucial for tracking the emergence and evolution of viruses over time. In environmental metagenomic surveillance, the samples are also complex, but in this case, with a major composition of unidentified microorganisms. The role of metagenomics in this context is to identify pathogens, investigate the pathogenic or virulence factors, and monitor changes and variations in the microorganisms present in the sample over time. Therefore, metagenomics in this context needs to be enhanced to extract the key features from such a complex and diverse sample and provide a “de novo” identification of uncharacterised pathogens. The use of metagenomic analysis is now also considered for performing the **detection of non-clinical human pathogens,** which involves the routine collection of samples from healthy individuals that volunteer at airports, train stations or military bases, a notable example being the CDC genomic surveillance program (CDC, 2024).

By encompassing diverse sample origins – clinical, environmental, and non-clinical human – metagenomics offers a comprehensive approach to pathogen detection. This approach transcends the limitations of traditional diagnostic methods, facilitating both the precise mapping of known pathogens and the discovery of novel ones, ultimately contributing to a more proactive and robust infectious disease surveillance framework.

Table 1. *The different sample origins in metagenomics analysis*

	Clinical samples	Environmental samples	Non-clinical human samples
Description	Collected for diagnosis and treatment of diseases from individuals exposed to a health threat or presenting specific symptoms	Collected from various environments (soil, water, air, etc.) for research and monitoring	Collected from asymptomatic individuals for monitoring and research purposes
Challenges	Ethical considerations, privacy concerns, potential for contamination	Sample heterogeneity and difficulty in obtaining representative samples	Ethical considerations, privacy concerns, logistic challenges in sample collection
Benefits of AI	Faster and more accurate diagnosis, identification of novel pathogens, personalized treatment, prediction of disease outbreaks	Rapid detection of environmental changes, identification of novel microbes, monitoring of ecosystem health, prediction of environmental risks	Early detection of emerging pathogens, monitoring of disease spread in communities, identification of risk factors for disease

Questions asked	• What pathogens are present in the sample? • What is the patient's microbiome composition? • What are the antimicrobial resistance patterns of the pathogens?	• What is the microbial diversity and composition in this environment? • What are the functional capabilities of the microbes present? • What are the antimicrobial resistance patterns of the pathogens?	• What pathogens are circulating in the community? • Are there any emerging or re-emerging pathogens? • How can we monitor the spread of diseases in real-time?

Source: *author's elaboration based on literature review findings.*

Moreover, metagenomic analysis involves multiple steps, and there is no universal workflow that can be applied to every study. In general, there are two main approaches to process metagenomic reads: mapping reads to existing reference genomes or assembling the genomes directly from the metagenomic data (Ghurye, Cepeda-Espinoza, & Pop, 2016). The latter approach is essential for the identification of novel pathogens for which the scientific community do not have reference genomes yet.

For this study, all types of scenarios, methodologies and approaches in metagenomics have been considered to assess their role in health preparedness. There are six main recurring steps that are very similar across different metagenomic research studies (Krause, et al., 2021b). These are described below:

- **Sample collection**: metagenomic studies use non-selective sampling methods to capture the overall microbial diversity. This step includes sample preparation and DNA extraction.
- **Sequencing, data preparation and preprocessing**: this step consists of the generation and preparation of the raw sequence data and the preprocessing, which includes data ingestion, format conversion, and quality control processes.
- **Aggregation analysis**: where the generated sequences are grouped, sorted and assembled as necessary for the analysis step. This step will vary depending on the type of input sequences, whether these come from targeted amplicon or whole-genome sequencing.
- **Feature engineering**: the objective of this step is to create the sequence features that will be later analysed. This step includes the annotation of the unknown sequences, mapping the pre-processed samples to reference genomes, or the prediction of specific genes and their functionalities. This step may also include processes to simplify the data (computation of relative abundances of genes, reduction of data complexity, etc) to facilitate the posterior analysis.
- **Analysis:** this step consists of the analysis of the generated features, new information is obtained from the input data. This is the more heterogeneous step as it's the one that specifically responds to the individual needs and questions of each research study. In general terms, this step usually consists of the identification and classification of the phenotypes and the computation of correlations between the identified features.
- **Data preparation for interaction and perception**: the final step of the workflow involves the preparation of the results in an informative way for the user to interpret them. This may include the construction of phylogenetic trees, clustering visualisation and correlation networks.

1.4.2 AI-based tools for the early detection of emerging pathogens via metagenomic sequencing

Unlike common bioinformatic classifiers designed for known pathogens, metagenomic sequencing necessitates additional algorithms specifically tailored for discovery, along with rigorous validation to assess performance in identifying new organisms (Carbo, et al., 2020).

Artificial intelligence (AI) is a rapidly evolving tool revolutionising many aspects of healthcare. It is a vast field that includes a variety of methodologies, such as computer vision, natural language processing, and deep learning. These methods are useful for pattern recognition, classification, and large-scale data analysis.

In 2023, OECD member countries approved a revised version of the organisation's definition of an AI system (OECD, Artificial intelligence, 2023). The updated definition reflects developments over the last five years, improves technical accuracy, and makes it more 'future-proof'. The EU's AI Act has aligned the definition of AI to the updated OECD definition in its final text. In this study, we will consider the updated OECD definition of 'AI systems' as a baseline to ensure technical accuracy and international alignment. The definition can be found in the box below.

Box 1. *Working definition of "AI system"*

> "An AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment."

The OECD working definition provided the overarching theoretical framework, but for the purposes of this study, we focused on specific AI techniques and algorithms. Particularly, the OECD Framework for the Classification of AI Systems provides a comprehensive and structured approach that helps characterise AI systems based on key dimensions such as description, examples, data and main tasks (OECD, 2022). The framework provided by the OECD has been used as a guide to establish a classification model by providing a systematic way to assess and categorise different AI systems based on their characteristics and the way that they produce outputs. As a result, the study team considered the following AI techniques, namely **Symbolic AI, Statistical AI, and Hybrid AI**. The Table below provides an overview of the three categories considered with their main characteristics and tasks performed. **Symbolic AI models** allow us to make inferences and perform classification (e.g., Knowledge-based and expert systems). **Statistical AI models** facilitate trend and pattern prediction based on data (e.g., Linear regression, Logistic regression, Decision trees, Neural networks, Naïve Bayes, Deep learning). Lastly, **Hybrid AI models** combine the previous approaches to extract complex features, enhance pattern recognition, and learning abilities.

Table 2. *Definition of the AI techniques considered in this study*

	Symbolic AI model	Statistical AI model	Hybrid AI model
Description	Symbolic or knowledge-based AI uses human-generated logical representations to infer a conclusion from a set of constraints (variables). These constraints include rules, ontologies and search algorithms that rely on explicit descriptions of variables and descriptions of the interrelations between these variables.	Statistical AI models (e.g. genetic algorithms, neural networks and deep learning) identify patterns based on data rather than expert human knowledge. Primarily used for recognition, more recently, these models are also being used for tasks such as generation, synthesising and generating images or audio.	Many applied AI systems combine symbolic and statistical models into "hybrid" models. For example, NLP algorithms often combine statistical approaches building on large amounts of data and symbolic approaches that consider issues such as grammar rules.

Examples	Knowledge-based and expert systems	Linear regression, Logistic regression, Decision trees, Neural networks, Naïve Bayes, Deep learning	NLP
Data	Labelled and categorised	Can be labelled or unlabelled	Can be labelled or unlabelled
Task	Making inferences and performing classification	Trend/ pattern prediction	Complex feature extraction and pattern recognition/ learning abilities

Source: *author's elaboration based on literature review findings.*

Prior to 2019, AI has been predominantly employed in medicine and healthcare administration, however, the COVID-19 pandemic shifted its focus from medicine to public health (Olawade, 2023). AI advancements are expected to play a key role in event-based surveillance, which is an important component of pandemic preparedness (Chiu & Miller, 2019), helping to identify areas of high risk for disease transmission between humans and animals, potentially preventing future pandemics. Current monitoring activities mainly focus on known diseases, but the rise of emerging diseases highlights the need for early detection of unusual illnesses or pathogens before they cause widespread human disease (Chiu, Sridhar, & Yuen, 2023). This early detection could significantly reduce the impact of future pandemics, especially considering the complex interplay between microbes and the host's immune system, which often makes it challenging to determine if a detected microorganism is harmful or harmless (Chiu & Miller, 2019).

During the COVID-19 pandemic, **AI early warning systems** became highly relevant and a substantial investment both in the EU and in other countries, which made it possible to advance AI tools for epidemic monitoring (MacIntyre, et al., 2023). The advantages of AI in early warning systems are multi-fold. It is rapid and real-time because the system generates epidemic alerts through the processing of open-source data online all over the world. It requires minimal human intervention and, therefore can address the issue of lack of expertise in the field theoretically (Chiu, Sridhar, & Yuen, 2023). **AI has also been adopted in metagenomics**, proving to be a game-changer and offering a powerful means to manage, process, and interpret the colossal amount of data generated. In metagenomic studies, the sheer volume of data is overwhelming - about millions and often billions of DNA sequences representing diverse and unidentified or unknown species. Sorting through this vast data repository and making sense of it is a daunting challenge. This is where AI steps in, enabling researchers to find patterns and make predictions in ways that would be otherwise impossible (Pandey, 2023).

1.4.3 Initial bottlenecks identified

Despite the potential and recent successes of metagenomics, its application in clinical practice and monitoring has lagged research advances owing to a number of factors.

Carbo et al. (2020) indicated that metagenomic analysis requires efficient DNA extraction protocols, computational tools, and annotated reference genome databases that are not always available. Each of these factors impacts metagenomics analysis to varying degrees. The Study will further examine these bottlenecks in Section 2.2. In parallel, the production of data has dramatically increased over the past years due to the decreased cost and improved efficiency of sequencing technologies. This spike in data generation has made the processing and analysis of these large datasets difficult. Limiting factors are partly the availability of computational resources but mainly the bioinformatics expertise in establishing and applying appropriate processing and analysis pipelines (Vollmers, Wiegand, & Kaster, 2017). Another relevant challenge at the human level is the need for biologists and biotechnologists to acquaint themselves with proficient data handling, digital microbiology, and the interpretation of the

working and outputs of AI systems. To tackle these issues, increasing efforts have been made at the global and EU levels, as detailed in the Overview of the state-of-the-art subsection.

This study contributes to gathering information on the issue by further exploring the current most pressing bottlenecks in metagenomics for emerging pathogen identification and its application in health preparedness and emergency response. Furthermore, recommendations on the use of AI tools for metagenomics analysis will be discussed in the Results chapter below.

2. Results

This section summarises the main results of the literature review and the enrichment of these with the information gathered in consultation activities with key experts in the area.

The first part focuses on the set of identified AI tools for improving metagenomic analysis, while the second part focuses on the remaining challenges of metagenomics, where innovation is still needed, and whether AI tools could be of help.

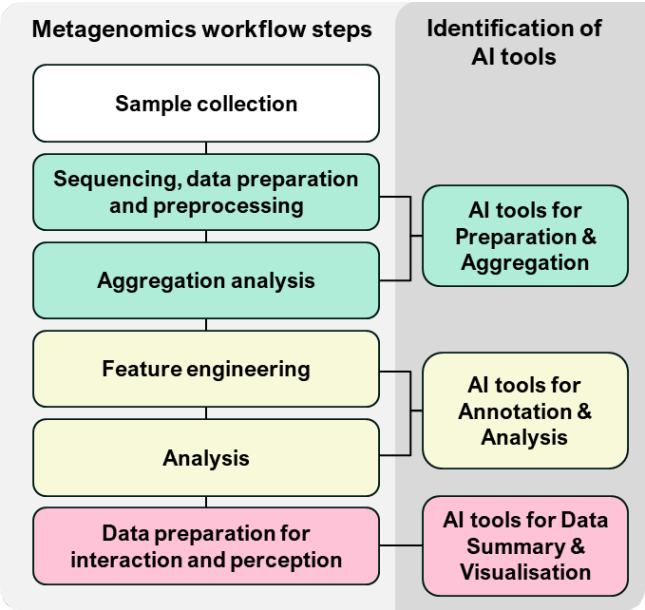
2.1 Use of AI-based tools for pathogen identification using metagenomic analyses

Metagenomic research studies often consist of six general steps that are very similar across studies (Krause, et al., 2021b). The six steps of a generic metagenomic workflow, as detailed in the overview of this report are: (1) Sample collection; (2) Sequencing, data preparation and preprocessing; (3) Aggregation analysis; (4) Feature engineering; (5) Analysis; and (6) Data preparation for interaction and perception.

For the purpose of this research, and after a thorough review of the literature, **three main phases of innovation in the metagenomic workflow have been defined**: **AI tools for data processing and aggregation** (covering steps 2 and 3 of the metagenomics workflow), **AI tools for annotation and analysis** (covering steps 4 and 5 of the metagenomics workflow), and **AI tools for data preparation for interaction and perception** (corresponding to step 6 of the metagenomics workflow). See the representation in **Error! Reference source not found.**.

In this report, we propose utilising the three identified phases of the workflow as a framework for identifying, evaluating and mapping the different AI-based tools tailored for metagenomics research. By aligning the tools with the metagenomics workflow, we seek to find the best tools to **improve the analysis of metagenomic data** at the different phases of the workflow.

***Figure 2**. Use of AI tools in the metagenomics workflow*



***Source**: author's elaboration based on literature review findings*

A key aspect of metagenomic studies is the complexity and multi-dimensional nature of the data. This data involves many features or variables, each representing a separate "dimension," such as different microbial species, genes, or metabolic pathways. This complexity requires accurate analytical tools, as the benefits offered by metagenomics are highly dependent on the efficient processing of the large amount of generated data. In this regard, Phase 1 tools are key for the preparation of raw and processed sequences for the next steps of a study. AI tools in this phase are crucial to ensure accuracy and good data quality, an essential step to obtain accurate results in downstream analyses. Moreover, these AI-based tools can optimise the data transferability and use of computational resources. These tools provide a preliminary identification of the type of pathogen through aggregation and mapping against reference genome datasets. Phase 2 AI-based tools are those that contribute to the feature extraction and analysis itself. These tools are key for a better understanding of the pathogenicity of the microorganisms present in a sample. These tools are also useful for measuring evolution rates, which is essential information in the early detection of outbreaks and in systematic surveillance. Last, phase 3 AI-based tools aim to facilitate the metagenomic results' interpret. A key aspect to ensure understanding of the obtained results by non-specialised bioinformaticians, allowing for a faster and more accurate response in case of emergency.

AI-based tools can be leveraged to help with the increased use of sequencing technology in the biomedical field and the enormous quantity of associated data. Building on the points made above, the following table outlines **the three phases in the metagenomics workflow and the benefits each AI technique brings to the process**. Symbolic AI models are highly compatible with the metagenomic workflow with respect to annotation and analysis, as they offer benefits in terms of increased accuracy and interpretability of results. By using human encoded knowledge and logical reasoning, it is a valuable tool for achieving reliable and understandable outcomes. Statistical AI models perform well within the metagenomics workflow with aggregation, as they can analyse large amounts of data that identify patterns and generate hypotheses. This AI technique effectively uncovers hidden patterns and generates hypotheses that lead to informed decision-making. Lastly, the hybrid AI model can perform multiple tasks such as event detection, forecasting, personalisation, reasoning with knowledge structures, and interaction support that ultimately proves to be effective for handling complexity and learning from raw data throughout the

metagenomics workflow. Additionally, due to these benefits, the hybrid AI's capabilities align well with the complexity and learning requirements of the metagenomics workflow, making it a suitable choice for managing large amounts of data.

Table 3. *Definition of the AI techniques considered in this study*

	Symbolic AI model	Statistical AI model	Hybrid AI model
Stage in the metagenomics workflow	Aggregation	Annotation & analysis	Several stages
Main benefits brought to the metagenomics workflow	Increased accuracy & interpretability of the results	• Discovering hidden patterns • Hypothesis generation	• Handling complexity • Learning from raw data

Source*: author's elaboration based on literature review findings*

The general use of AI in metagenomic studies for health preparedness and the approach used to map these tools was discussed during the consultation activities of the experts. The main highlights resulting from these discussions can be found in **Error! Reference source not found.**. A detailed description of the use of AI in the different phases of metagenomic studies can be found in the following sections below.

Box 2. *Considerations from the experts-Use of AI-based tools for pathogen identification using metagenomic analysis*

Highlights resulting from Experts' consultation

One stakeholder with experience both in Academia and the Industry shared that the **promise of AI for metagenomics is substantial, especially in terms of reducing the time and computational resources required to process sequencing data**. Considering that newer sequencing technologies tend to generate larger datasets, AI's ability to efficiently process large amounts of data is crucial.

Regarding the type of AI techniques or methods considered as most promising for metagenomic analysis, experts from both academia and the industry noted that **deep learning techniques stand out as one of the most promising AI methods**. These techniques outperform traditional bioinformatic tools in the identification of complex patterns in vast datasets. **Other experts recalled the importance of Natural Language Processing (NLP)** approaches to perform encoding, decoding and transforming tasks which are often more useful than specific tools used for specific steps in a metagenomic workflow.

Experts were asked to validate the approach followed for the mapping of AI-based tools for enhancing metagenomic workflows. In general, all **experts agreed with the proposed classification approach and some of them contributed with some additional tools and techniques** (for details on the suggested tools for consideration, see the sections below).

Several experts from academia reflected on the **importance of considering the source of the sample to be analysed in a metagenomic study**. Experts emphasised the need to differentiate between clinical samples and those from environmental or human non-clinical monitoring. In clinical settings, the focus is typically on mapping human samples against known pathogens. However, in wastewater surveillance (environmental surveillance), researchers often explore samples to uncover unknown organisms or track the exponential growth of specific microbes. Metagenomic surveillance goes beyond identifying previously known

pathogen sequences; it also involves identifying sequences that have not yet been identified or included in reference databases and monitoring through time genetic variations that could increase pathogenicity in human population. Therefore, **AI's role in surveillance is particularly beneficial compared to its impact on more mature and standardised areas like clinical metagenomics**.

Some experts with a background in public health laboratories provided further **insights into the opportunities posed by AI in the implementation of metagenomics in clinical practice**. Experts in clinical metagenomics noted that the implementation of large-scale metagenomic sequencing in clinical practice is very challenging. AI could aid by prioritising which patients' data sampled should undergo also metagenomic analysis, optimising the use of resources. Moreover, AI-based tools, through the identification of complex patterns, could improve the accuracy of results in clinical metagenomics by better differentiating between true pathogens and the wide variety of existing contaminants in clinical samples. However, there are also experts in clinical metagenomics who prefer traditional bioinformatic tools, as AI-based tools can sometimes be too sensitive. It was shared as an example of the outperformance of traditional bioinformatic tools in the identification of antibiotic resistance genes.

In metagenomic monitoring activities, some researchers believe that the most promising application of AI is in **bridging the gap between vast amounts of genotype and phenotype data.** It also highlighted the role that AI could play in the generation of FASTQ files and in the identification of pathogens in very complex samples. For the latter, it was highlighted by several experts the **importance of AI, especially for metagenomic surveillance, to identify pathogens which aren't represented in databases of well-known microorganisms**.

2.1.1 AI tools for the preparation and aggregation of the genomic sequences in metagenomic studies

These are the main takeaways on the role of AI-based tools enhancing the preprocessing and assembly of metagenomic sequences.

- Sample and sequence preparation is a key initial step in every metagenomic study. **AI tools, especially deep learning techniques, offer significant advances in the accuracy of base calling of various sequencing technologies**. This increase in accuracy reduces the time-consuming data curation process while increasing the accuracy of the entire metagenomic workflow.
- **Clustering and dimensionality reduction AI algorithms, despite being simpler algorithms, play a crucial role in the aggregation of sequencing data**, leading to a smoother analytic phase. Read binning is also facilitated using AI tools and their ability to recognise complex patterns in the sequences analysed, improving the efficiency of the analysis.

Most of the identified tools in the literature are used for Operational Taxonomic Unit (OTU) clustering and Read Binning. However, innovative **AI-tools developed for improving the accuracy of base-calling of innovative sequencing technologies like Nanopore hold great promise as they boost the fast and portable new sequencing technologies**.

Following sample collection, **the first essential step of a metagenomic analysis is to prepare and preprocess the raw sequence data** using several techniques aimed at ensuring the quality, integrity, and usability of the genetic data obtained from the samples (Krause, et al., 2021b). These techniques include data ingestion, format conversion and sequence quality control, among others. This step is integral to subsequent analyses and interpretations. The literature analysis has provided an overview of the availability of AI-based tools for this phase, which is mostly focused on the improvement of base-calling of sequencing platforms.

As reported by Cacho et al. (2016) and Abdelwahab et al. (2023)**, several computationally efficient algorithms have been developed to improve the accuracy and speed of base-calling**. Base-calling is the initial step in any sequencing methodology, where nucleotides are assigned based on the sequencing signal.

Illumina NGS (Illumina, Inc.) is one of the most widely used sequencing platforms that, by leveraging sequencing by synthesis (SBS) technology, proved very accurate at short-read sequencing. Some of the tools identified in the literature are **AltaCyclic** (Erlich, Mitra, delaBastide, McCombie, & Hannon, 2008), **Ibis** (Kircher, Stenzel, & Kelso, 2009) and **freelbis** (Renaud, Kircher, Stenzel, & Kelso, 2013). These tools, based on **support vector machines, are used to optimise the base-calling of Illumina Sequencing technologies**.

Sequencing technologies are improving fast, and in the last two decades, alternative sequencing platforms producing longer reads (PacBio, Oxford Nanopore Technologies) have acquired more relevance in the field. Nanopore sequencing technology (Oxford Nanopore Technologies Ltd.) is a type of long-read sequencing technique that, by translocating DNA fragments through an array of nanopores, sequences samples in a wide spectrum, focusing especially on ultralong reads. Longer reads improve the accuracy of alignment in repetitive regions; however, these are known for being less accurate in base calling. Besides the advantages and limitations of longer reads, Nanopore sequencing technologies offer real-time and in-field analysis of the samples, a characteristic that allows the sequencing of samples directly at the point of care (Wang, Zhao, Bollas, Wang, & Au, 2021). The literature search identified **several tools based on AI Deep Learning techniques for the preprocessing of Nanopore raw signals into nucleotide sequences**. Among these tools, there are **Chiron** (Teng, et al., 2018), **NanoDeep** (Lin, et al., 2024), and **Fast-Bonito** (Xu, et al., 2021). On the one hand, Chiron and Fast-bonito are open-source Deep Learning models that translate nanopore raw signals directly into nucleotide sequences, allowing for better accuracy in base-calling without losing speed and without specialised computer resources. On the other hand, NanoDeep is a deep learning framework that incorporates a convolutional neural network in the process, allowing rapid operations in parallel at read-time sequencing, exhibiting a better performance in speed and accuracy than the existing deep learning model (Yusen, et al., 2024). The main limitations of sequencing techniques providing longer reads are the requirement for relatively high amounts of nucleic acid material and, more concerning in the case of Nanopore sequencing, the relatively high error rates (Wang, Zhao, Bollas, Wang, & Au, 2021). The results of the literature review show how the **implementation of AI tools can improve the accuracy of sequencing technologies generating longer reads while maintaining the computational resources and the time of processing that make this type of resource so valuable, especially for in-site analysis of complex samples**.

The field of sequencing technologies is continuously evolving, and recent data point to the advantages of using a combination of short and long read sequencing technologies (Abdelwahab, Belzile, & Torkamaneh, 2023) (Freed, et al., 2022). The literature review has identified two **AI-based variant calling tools that can call variants simultaneously from a combination of short and long reads: DNAscope** (Freed, et al., 2022), a tool based on supervised learning techniques with Gradient Boosting Machines, and **DeepVariant** (Supernat, Vidarsson, Steen, & Stokowy, 2018), a tool based on deep neural networks. These methods show improvements in both variant calling and genotyping accuracy while keeping high accuracy even at lower coverages compared to standard variant calling tools (Freed, et

al., 2022). However, these variant callers were designed for human genomes, having a lower impact in enhancing metagenomic pathogen analysis.

After the preprocessing of the raw sequence data, **the next step is the aggregation of the individual sequences**. In this phase, the steps may differ depending on the sequencing technique used.

For sequences resulting from targeted amplicon sequencing, the aggregation phase primarily consists of OTU Clustering, grouping sequences according to their similarity to identify those belonging to the same or closely related taxonomic clades using the so called Operational Taxonomic Units (OTU) (Krause, et al., 2021a). Moreover, more attention is currently being drawn to the assignment of Amplicon Sequence Variant (ASV), which allow a finer distinction between sequences analysed (Callahan, McMurdie, & Holmes, 2017). Similar clustering techniques are often used for the assignment of both OTUs and ASVs. Clustering is a well-known machine learning technique, and **several clustering algorithms have been reported in the literature for their contribution to OTU clustering**. An example of a tool using clustering algorithms is **UPARSE-OTU** (Edgar R. C., 2013). UPARSE-OTU is a greedy algorithm (see Glossary) that performs chimera filtering and clustering simultaneously, assigning OTU clusters in a single iteration where similarity is based on the maximum parsimony model. This combination improves the accuracy of the OTU clustering (Edgar R. C., 2013). Greedy clustering algorithms are commonly used for OTU clustering for their speed and scalability and can be found in other tools like **Cd-hit** (Li & Godzik, 2006), **DySC** (Zheng, Kramer, & Schmidt, 2012), USEARCH – **UCLUST** algorithm (Edgar R. , 2010), **DNAClust** (Ghodsi, Liu, & Pop, 2011), **GramCluster** (Russell, Way, Benson, & Sayood, 2010), **VSEARCH** (Rognes, Flouri, Nichols, Quince, & Mahé, 2016), **SUMACLUST** (Mercier, Boyer, Bonin, & Coissac, 2013)(Mercier et al. 2013), and **OTUCLUST** in MICCA (Albanese, Fontana, Filippo, Cavalieri, & Donati, 2015).

As can be observed in the paragraph above, many tools use greedy clustering for the aggregation of individual sequences. This is mainly due to the reduced computational demands and memory usage of these methods, making them highly suitable for processing the large datasets generated by modern sequencing technologies. However, while these heuristic methods excel in speed and scalability, they often produce less accurate clustering results compared to hierarchical methods based on pairwise distance matrices (Wei et al., 2021). Alignment-based clustering methods like **DOTUR** (Schloss, Patrick, & and Handelsman, 2005) and **Mothur** (Schloss, et al., 2009), despite their higher accuracy, are extremely computationally complex due to the need of storing a complete pairwise distance matrix. In fact, when there are sequencing errors in the sample, these methods tend to overestimate the number of OTUs, making it problematic to be used for OTU clustering of large sequence datasets (Soueidan & Nikolski, 2016). Some initiatives explored the use of alternative methods to improve the scalability of hierarchical clustering methods. An example is **ESPRIT-Forest** (Cai, et al., 2017), an algorithm that allows the parallel hierarchical clustering of sequences, achieving subquadratic time and space complexity while maintaining a high OTU clustering accuracy. In general, a **trade-off is observed between the quality of the OTU clustering and the time and computational complexity of the methods** (Bhat, Prabhu, & Balakrishnan, 2019).

To overcome the limitations of the heuristic and hierarchical clustering methods, some research lines have focused on the development of model-based clustering methods. For example, the Bayesian clustering **CROP** (Hao, Jiang, & Chen, 2011) implements an unsupervised Bayesian clustering method to find clusters without a predefined threshold, improving its accuracy and reducing potential biases. Other examples of tools based on Bayesian clustering are **BACDNAS** (Jaaskinen, Parkkinen, Cheng, & Corander, 2013) and **BEBaC** (Cheng, Walker, & Corander, 2012). Other lines of research have focused on network based models like **DMclust** (Wei, Zhang, & Zhang, 2017) or **M-pick** (Wang, Yao, Sun, & Mai, 2013), a modularity-based clustering method for OTU picking which doesn't require predefined distance levels to define the OTUs.

For sequences resulting from whole genome sequencing, Read Binning is used to separate reads from different organisms either before initiating the assembly or after a pre-assembly step has been performed. The metagenomic assembly is the alignment and organisation of the individual short reads into longer contiguous sequences called contigs (Krause, et al., 2021b). There are two types of binning methods - those that rely on a reference database and those that perform reference-free binning by using clustering techniques on the features of the pre-processed raw sequences like GC content and *k*-mer frequencies. The application of AI techniques is more relevant in the latter, where there isn't a reference database to map the short reads. As in OTU clustering, **there are different clustering methods used to improve Read Binning**, and AI-techniques have been a key resource to improve unsupervised clustering methods. Regarding unsupervised clustering, there are several tools based on the k-mer distribution. Examples of **these tools based on k-means clustering** are **CompostBin** (Chatterji, Yamazaki, Bai, & Eisen, 2008), **MetaCluster** (Wang, Leung, Yiu, & Chin, 2012)**,** and **Tetra** (Teeling, Waldmann, Lombardot, Bauer, & Glockner, 2004). Within this group, some tools like Latent Strain Analysis **(LSA)** (Cleary, et al., 2015) have integrated additional ML techniques like dimensionality reduction algorithms and feature hashing to improve the clustering performance. Some other tools have been developed relying on model-based clustering to improve the performance of the clustering in samples with shorter reads, as in **SCIMM** (Kelley & Salzberg, 2010) and **LikelyBin** (Kislyuk, Bhatnagar, Dushoff, & and Weitz, 2009)**.**

All these clustering algorithms heavily rely on *k*-mer abundance to differentiate reads from different species. However, closely related species might have very similar *k*-mer abundances, affecting the accuracy of the clustering (Soueidan & Nikolski, 2016). Some tools have been developed to simultaneously analyse data from multiple samples to compensate for the *k*-mer abundances issue in the data from unique samples. For example, **CONCOCT** (Alneberg, et al., 2014) clusters the reads based on sequence composition but also considers the coverage across multiple samples using Gaussian mixture models.

Assembling genomes involves piecing together overlapping reads to reconstruct longer stretches of DNA. The goal of read assembly is to reconstruct as much of the original genetic material as possible, allowing for subsequent analysis, such as identifying genes, pathways, and species present in the sample. This study has identified several **tools based on AI that contribute to the accuracy and speed of the assembly of short sequencing reads**. **MetaVelvet** is a metagenomic short sequence read assembly tool based on a de Bruijn graph which has extensions like **MetaVelvet-SL,** which uses an SVM for learning the classification model based on 94 features extracted from candidate nodes (Afiahayati, K, & Y., 2015), and **MetaVelvet-DL,** which relies on Convolutional Neural Networks (Liang & Sakakibara, 2021).

The literature search also identified other tools for metagenomic assembly based on Deep Learning techniques. An example of this tool is **DeepMAsED**, a tool that accurately identifies misassemblies in metagenome-assembled contigs, demonstrating that **deep learning is a feasible solution for the detection of misassembly of metagenome-assembled contigs without the need for reference genomes**. DeepMAsED has been proven to be accurate even with samples including a diversity of bacteria and archaea without the need for strong modelling assumptions or reference genomes (Mineeva, Rojas-Carulla, Ley, Schölkopf, & Youngblut, 2020).

AI techniques offer significant advantages for the preparation, aggregation and assembly of sequences in metagenomic analysis. More specifically, ML methods, such as SVMs, can enhance the accuracy of read binning by learning complex patterns in sequence composition, coverage, and similarity. AI algorithms, particularly deep learning models, can also effectively handle high-dimensional, noisy data, enabling accurate assembly of fragmented reads into longer contigs.

Box 3. *Considerations from the experts: AI tools for the preparation and aggregation of the genomic sequences*

Highlights resulting from Experts' consultation

All experts agreed on the **importance of considering the source and characteristics of the sequence** to select the right bioinformatic tools and define the required preprocessing of the raw sequences**.** One industry expert noted that sample preparation is a key step in the processing of the samples. Depending on the origin of the samples, there are many decisions to be taken in relation to the need to enrich or remove certain genetic material, the type of the organisms targeted in the analysis (virus, bacteria), the type of genetic material (DNA, RNA), etc. These decisions could be better informed with the help of AI-based solutions.

Several experts from the Industry and Academia emphasised the **contribution of AI-based tools for the preprocessing and preparation of genomic sequences, especially those tools using Deep Learning**.

Experts coming from academia agreed that Nanopore sequencing holds great promise for metagenomic studies as it produces longer reads which reduce the ambiguity and potential artefacts introduced while aligning shorter reads into contigs. However, the base calling accuracy of this technology was an inhibiting shortcoming for the adoption of the ONT technology. Several of these experts noted the key role of several Deep Learning tools for the improvement in the processing of sequences resulting from Nanopore sequencing technologies. Recent **advances in AI have enhanced the accuracy of base calling, making Nanopore sequencing much more viable.** Experts provided some example data processing toolkits that contain Oxford Nanopore Technologies' base calling algorithms and several bioinformatic post-processing features like "Dorado" and "Guppy". The improvement in accuracy of Nanopore sequencing was highlighted as a significant change in industry and academia as it enables the use of an efficient method without losing accuracy.

Several experts from academia commented on the **use of clustering techniques, a simple form of Machine Learning, for the aggregation of sequence data**. There were various opinions on the relevance and usefulness of clustering techniques among the interviewed experts. Some of them noted that using these techniques has an important downside which is losing data features throughout the process. Other stakeholders with experience in national monitoring programmes noted that although these are very basic AI algorithms, these are still very relevant and widely used in the field.

Finally, some stakeholders from the industry noted that there are various **adapter removal tools (necessary for the preprocessing of sequences generated with NGS) that use different basic AI techniques**. These tools, although not using very advanced AI algorithms, allow the automation of the processing of large datasets by making inferences and statistical decisions.

The table below provides a quick overview of the main AI tools identified in the phase of metagenomic data annotation and analysis categorised by step, AI technique utilised, and the availability in the market. Tools that resulted from consultations with the experts have been integrated at a later stage.

***Table 4**. List of AI tools identified for the preparation and aggregation of the genomic sequences in metagenomic studies*

Application	Techniques	AI-based tools example and reference	Availability[a]
Raw data processing	Support vector machines	AltaCyclic (Erlich, Mitra, delaBastide, McCombie, & Hannon, 2008)	Open source
		Ibis (Kircher, Stenzel, & Kelso, 2009)	Open source
		freeIbis (Renaud, Kircher, Stenzel, & Kelso, 2013)	Open source
	Deep learning	Chiron (Teng, et al., 2018)	Open source
		NanoDeep (Lin, et al., 2024)	Open source
		Fast-Bonito (Xu, et al., 2021)	Open source
		DeepVariant (Supernat, Vidarsson, Steen, & Stokowy, 2018)	Open source
		Dorado (Seymour, s.d.)	Open source
		Guppy (Oxford Nanopore Technologies, s.d.)	Proprietary - ONT[b]
		Megalodon (Oxford Nanopore Technologies, s.d.)	Proprietary - ONT[b]
		DeepSignal (Ni, et al., 2019)	Open source
		DeepMod2 (Ahsan, Gouru, Chan, Zhou, & Wang, 2024)	Open source
		DeepMP (Bonet, et al., 2022)	Open source
		NanoCaller (Ahsan, Liu, Fang, & Wan, 2021)	Open source
		Medaka (Oxford Nanopore Technologies, s.d.)	Open source
	Gradient Boosting Machines	DNAscope (Freed, et al., 2022)	Open source
OUT Clustering	Heuristic clustering algorithms	UPARSE-OTU (Krause, et al., 2021b)	Open source
		Cd-hit (Li & Godzik, 2006)	Open source
		DySC (Zheng, Kramer, & Schmidt, 2012)	Open source
		UCLUST (Edgar R. , 2010)	Open Source
		DNAClust (Ghodsi, Liu, & Pop, 2011)	Open source
		GramCluster (Russell, Way, Benson, & Sayood, 2010)	Open source
		VSEARCH (Rognes, Flouri, Nichols, Quince, & Mahé, 2016)	Open source
		SUMACLUST (Mercier, Boyer, Bonin, & Coissac, 2013)	Open source
		OTUCLUST (Albanese, Fontana, Filippo, Cavalieri, & Donati, 2015)	Open source
	Hierarchical clustering algorithms	DOTUR (Schloss, Patrick, & and Handelsman, 2005)	Open source

		Mothur (Schloss, et al., 2009)	Open source
		ESPRIT-Forest (Cai, et al., 2017)	Open source
	Model based clustering algorithms (Bayesian modelling)	CROP (Hao, Jiang, & Chen, 2011)	Open source
		BACDNAS (Jaaskinen, Parkkinen, Cheng, & Corander, 2013)	Open source
		BEBaC (Cheng, Walker, & Corander, 2012)	Open source
	Network based clustering algorithms	M-pick (Wang, Yao, Sun, & Mai, 2013)	Open source
		DMclust (Wei, Zhang, & Zhang, 2017)	Open source
Read binning	Clustering algorithms (k-means clustering)	CompostBin (Chatterji, Yamazaki, Bai, & Eisen, 2008),	Open source
		MetaCluster (Wang, Leung, Yiu, & Chin, 2012)	Open source
		Tetra (Teeling, Waldmann, Lombardot, Bauer, & Glockner, 2004).	Open source
		Latent Strain Analysis (LSA) (Cleary, et al., 2015)	Open source
	Clustering algorithms (model based clustering)	SCIMM (Kelley & Salzberg, 2010)	Open source
		LikelyBin (Kislyuk, Bhatnagar, Dushoff, & and Weitz, 2009)	Open source
	Clustering algorithms	CONCOCT (Alneberg, et al., 2014)	Open source
Read assembly	Support Vector Machines	MetaVelvet-SL (Afiahayati, K, & Y., 2015)	Open source
	Deep learning	DeepMAsED (Mineeva, Rojas-Carulla, Ley, Schölkopf, & Youngblut, 2020)	Open source
		MetaVelvet-DL (Liang & Sakakibara, 2021)	Open source

***Source**: author's elaboration based on literature review findings.*

[a] *For this study, considering that the vast majority of tools is used for research purposes and in tight collaboration between the industry, academia and the clinical settings, the availability of the tool refers to the type of access to the source of the tool, distinguishing between "**open source**" and "**proprietary**", indicating the owner of the tool where possible.*

[b] *ONT (Oxford Nanopore Technologies)*

2.1.2 AI tools for the annotation and analysis of the genomic sequences

These are the main takeaways on the role of AI-based tools enhancing the annotation and analysis of metagenomic sequences.

- Most of the tools used are **deep learning** techniques.
- A key step in this phase of the metagenomic analysis is **gene prediction** which is facilitated using AI tools that improves gene prediction accuracy, especially for novel sequences with limited similarity to reference databases. AI tools are considered able to innovate the whole feature engineering process.

- The identification of these tools reflects the importance of AI in making metagenomic sequencing more accurate, especially when dealing with genes that do not have a known reference in the database.

A gap is observed in the use of AI tools for **diversity estimation.** During the interviews with experts, a bioinformatic tool, Nonpareil (Rodriguez-R. & Kostantinos, 2013), has been cited as useful during this step. However, no AI tools have been mentioned.

Once the sequencing reads have been aggregated and assembled to reconstruct as much of the original genetic material as possible, the next phase of the metagenomic workflow consists of the annotation and analysis of the Metagenome-Assembled Genomes (MAGs). This phase **systematically investigates the reconstructed genomes, elucidating the functional capacity and taxonomic structure of the original microbial community** (De Fátima Alves, et al., 2018).

Firstly, **specialised bioinformatic algorithms are used to scan the MAGs and predict potential coding regions**. Gene identification and annotation is a crucial step in microbiome analysis (Shen, et al., 2023) and the challenge of identifying genes within novel MAGs provides a context well-suited to **deep learning approaches**. Indeed, AI, and particularly deep learning, improves gene prediction accuracy, especially for novel sequences that lack close matches in reference databases. However, other AI techniques are also often applied to gene prediction, as models can learn complex patterns in DNA sequences that may not be obvious to traditional algorithms. For example, tools like **Orphelia** (Hoff, Lingner, Meinicke, & Tech, 2009) are great for *ab initio* gene prediction. The performance of Orphelia was compared against those of **MetaGene** (Noguchi, Park, & and Takagi, MetaGene: prokaryotic gene finding from environmental shotgun sequences, 2006) and its successor **MetaGeneAnnotator** (Noguchi, Taniguchi, & and Itoh, 2008) and the evaluation on simulated data demonstrated that Orphelia has a higher gene prediction accuracy**.** Deep learning allows models to learn gene properties through comprehensive data training, improving prediction performance. One such deep-learning method is **Meta-MFDL** (Yang et al., 2021; Krause et al., 2021a) combining multiple features for metagenomic gene prediction. In the deep learning field, Convolutional neural networks (CNNs) have also proven highly successful in metagenomic gene prediction tools. **CNN-MGP** (Al-Ajlan & El Allalli, 2018), a convolutional neural network-based metagenomics gene prediction tool, boasts over 90% accuracy in test datasets. CNN-MGP scripts are designed to predict protein-coding genes in short environmental DNA sequences with unknown phylogenetic origin.

Alternative specialised tools designed for phage gene annotation, such as **PHANOTATE** (McNair, Zhou, Dinsdale, Souza, & Edwards, 2019) are also important when dealing with their unique genomic structures. Existing gene annotation tools often struggle with the compact nature of phage genomes, where genes can overlap or exist entirely within other genes. PHANOTATE, the first phage-specific gene prediction tool (Shen, et al., 2023), designed by McNair et al. (2019) employs a novel approach specifically for phage genomes, treating them as networks of paths and utilising dynamic programming to find the optimal gene prediction path. In short-read DNA sequencing experiments with very low coverage, the original sequence reconstruction from the reads can be difficult because of the occurrence of uncovered gaps. AI can be of great help in this instance, as it excels at complex pattern recognition.

Next, these predicted genes are **compared to curated gene databases to identify already known genes (map to gene database) and assign them potential functions (functional annotation**), based on their similarities. As part of the annotation process, researchers aim to identify and characterise the genetic sequences present in their sample, indeed, functional annotation tries to answer the question of “what do the components do?”. This functional annotation serves as a foundation for subsequent analyses, including phenotype classification, diversity assessment, and phylogenetic reconstruction. Traditional homology-based methods compare DNA sequences to known genes by identifying homologs with known functions;

researchers can then assign putative functions to these DNA sequences. A major limitation of these methods is the absence of well-characterised sequences for certain functions (Sinha, Lynn, & Desai, 2020). **AI-based methods,** on the other hand, find patterns in the DNA that traditional methods might miss. AI techniques enhance functional annotation through better gene function prediction and by identifying potential functions in sequences with no known homologs. Deep learning methods have significantly improved the ability to predict gene functions, even for sequences that lack clear similarities to well-characterised genes in databases. Tools like **MetaMLP** (Arango-Argoty, Heath, Pruden, Vikesland, & Zhang, 2019) or **DeepFRI** (Gligorijević, et al., 2021) utilise sequence similarity searches against protein databases for functional annotation of unknown sequences in a metagenomic sample. In this step, the literature reviewed only showed evidence of deep learning techniques being used. One reason could be that clustering algorithms or other unsupervised learning techniques do not differ much from bioinformatic tools already being used, one notable example being Foldseek (van Kempen, Kim, Tumescheit, & al., 2024).

There are examples of Random Forest, **PaPrBaG** (Deneke, Rentzsch, & & Renard, PaPrBaG: A machine learning approach for the detection of novel pathogens from NGS data., 2017) a tool that overcomes genetic divergence by training on a wide range of species with known pathogenicity phenotype and remains reliable even at low coverage. Examples of logistic regression tools were provided by Soueidan & Nikolski (2016) in their review of machine learning tools for metagenomics, for example: **MetaPhyl**. Other logistic regression tools are **HyperVR** (Ji, et al., 2023), a hybrid deep ensemble learning approach for simultaneously predicting virulence factors and antibiotic resistance genes, and **PlasClass** (Pellow, Mizrahi, & Shamir, 2020), a new plasmid classifier. PlasClass outperforms the state-of-the-art plasmid classification tool on contigs assembled from shorter sequences, allowing it to achieve higher F1 scores in classifying sequences from a wide range of datasets. Most importantly, PlasClass also uses significantly less time and memory.

Once the genes have been identified and their functions classified through functional annotation, researchers classify the different organisms within the sample (**taxonomic annotation**). While recent taxonomic classification programs achieve high speed by comparing k-mers, they often lack sensitivity for overcoming evolutionary divergence, meaning that large fractions of the metagenomic reads remain unclassified. AI-driven models can improve identification accuracy, especially for organisms with less representation in reference databases. By using the presence and absence of specific genes, it is possible to identify the taxonomic groups represented in the MAGs and, thus, the organism or group they belong to (Park, et al., 2023).

Park et al. (2023) report that AI tools aid in taxonomic classification by learning complex patterns that differentiate microorganisms, an example being **DeepMicrobes,** a deep learning framework utilising bidirectional long short-term memory (LSTM) and a self-attention mechanism for classifying metagenomic reads at the species and genus level. According to Shen et al. (2023), **GOTTCHA** (Genomic Origins Through Taxonomic CHAllenge) is an application of a novel, gene-independent and signature-based metagenomic taxonomic profiling method with significantly smaller false discovery rates (FDR) and is also laptop deployable. Most importantly, it supports long reads. **Bracken** (Bayesian Re-estimation of Abundance with KrakEN) (Lu, Breitwieser, Thielen, & Salzberg, 2017), also cited both in Saha et al. (2020) and Jiang et al. (2023), is a tool meant to enhance the results of Kraken (Lu, Rincon, Wood, & al., 2022), another taxonomic classification tool. **IDTAXA** (Murali, Bhargava, & Wright, 2018) instead improves the accuracy of taxonomic classification by reducing excessive classification errors.

Gene prediction, functional annotation, and taxonomic classification form the core of the analysis of the MAGs. Following these steps, the focus shifts to transformational tasks such as the computation of relative abundances, where the previously identified genes or taxa are counted and put into relation to other genes or taxa in the same sample, or other more complex transformations known as “feature selection” and “feature extraction” (Krause, et al., 2021a).

In fact, researchers apply various feature selection and extraction techniques to remove irrelevant and redundant features from high-dimensional metagenomic datasets (Krause, et al., 2021b). The literature review revealed limited integration of AI, specifically within the feature selection step. Identified tools emerged largely from recent publications, with the award-winning example being **Xtreme Gradient Boosting (XGBoost)** (Krause, et al., 2021b). In fact, Xtreme Gradient Boosting was not created specifically for the metagenomics workflow, but its higher accuracy than other single decision tree algorithms, made it a suitable tool to be employed in metagenomics. Moreover, Wassan et al. (2019) conducted a comprehensive analysis of metagenomic data from human microbiomes with the application of different feature selection strategies and found that the application of embedded feature selection methods, namely extreme gradient boosting (XGBoost), led to better results than logistic regression techniques when determining important microbial genes in metagenomic studies.

Following feature engineering, analysis techniques generate valuable insights from the data. Such analyses include diversity estimation, phylogenetic analysis, phenotype classification and correlation analysis. At this stage, the analysis of the literature did not find an abundance of AI tools, apart from those used for phenotype classification. Below are detailed all the steps of the metagenomics workflow together with a description of the AI-based tools identified in the literature for each step.

Diversity estimation in metagenomics is peculiar to the amplicon workflow, and it focuses on characterising the variety and distribution of organisms within a microbial community (Rasheed, Rangwala, & Barbará, 2013). It could be either taxonomic-based (relies on assigning sequences to taxonomic groups and calculating diversity indices) or phylogenetic-based (accounts for evolutionary relationships between organisms using metrics like UniFrac distance). **Phylogenetic analysis** is also peculiar to the amplicon workflow. In metagenomics, researchers use phylogenetic trees to visualise these relationships based on the similarity of marker genes (e.g., 16S rRNA gene for bacteria and archaea).

Evidence showed the use of AI to help with **phenotype classification,** which is used to predict phenotypes (observable traits) of microbes within a community. This can include aspects like metabolic capabilities and antibiotic resistance potential. Literature shows that the AI tools employed at this step of the analysis are mainly at a research/ development stage and deep learning techniques. The only exception found is the Support Vector Machine **VirulentPred** (Ji, et al., 2023), which was developed for the purpose of detecting virulence factors (VFs) and combines various feature engineering techniques to generate inputs for distinct machine learning classification models. Examples of deep learning techniques include **DeePaC** (Johnson, Vinatzer, & Li, 2023; Bartoszewicz, Seidel, Rentzsch, & Renard, 2020), **DeepTE** (Yan, Bombarely, & Li, 2020) **DeepForest** (Zhou & Feng, 2017) and **DeepARG**, a deep learning approach for predicting antibiotic resistance genes from metagenomic data (Arango-Argoty, Heath, Pruden, Vikesland, & Zhang, 2019), virulence factors or pathogenicity. **DCiPatho** (Jiang, et al., 2023) has been shown to utilise Neural Networks to obtain the prediction of pathogenic probabilities, starting from known pathogenic and non-pathogenic sequences. The open tool **BacPaCS** (Barash E. , Sal-Man, Sabato, & Ziv-Ukelson, 2019) is a support vector machine (SVM) used for multi-class classification of microorganisms with metagenomic sequence datasets. The BacPaCS tool specifically handles datasets significantly larger than those previously used and shows high accuracy in distinguishing pathogenic from non-pathogenic bacteria in a clinically relevant dataset comprising only human-hosted bacteria.

Finally, the annotation and analysis phase of the workflow is completed by correlation analysis, which explores the statistical relationships between different variables within metagenomic data.

Box 4. *Considerations from the experts: AI tools for the annotation and analysis of the genomic sequences*

Highlights resulting from Experts' consultation

When presented with the matrix of AI tools resulting from the literature review conducted, the experts' consultations overall **confirmed the findings of the literature review**, highlighting especially that **Kraken and Bracken** have proven useful in the work of clinical researchers.

One stakeholder (Academia) identified the **taxonomic annotation** step as extremely important both for wastewater surveillance and detection of pathogens. It is worth citing that GeNet and BERTax have been cited as useful in the work of the stakeholder in the taxonomic annotation step.

Another stakeholder (Academia) considered the whole **feature engineering** stage (thus including gene prediction, mapping to gene databases, functional annotation, taxonomic annotation and feature selection) as one of the areas or processes in metagenomics that could **benefit the most from AI innovation**. Literature review findings were scarce regarding AI tools at the Phylogenetic analysis phase, but one stakeholder cited UShER.

The table below provides a quick overview of the main AI tools identified in the phase of metagenomic data annotation and analysis categorised by step, the AI technique utilised, and the availability in the market. Tools that resulted from consultations with the experts have been integrated at a later stage.

Table 5*. List of AI tools identified for the annotation and analysis of the genomic sequences*

Application	Techniques	AI-based tools example and reference	Availability[a]
Gene prediction	Random forest	PaPrBaG (Johnson, Vinatzer, & Li, 2023; Deneke, Rentzsch, & Renard, 2017)	Open source
	Logistic regression	PlasClass (Pellow, Mizrahi, & Shamir, 2020; Jiang, et al., 2023)	Open source
		MetaGene (Noguchi, Park, & and Takagi, 2006)	Open source
		MetaGeneAnnotator (Noguchi, Taniguchi, & and Itoh, 2008)	Open source
		MetaPhyl (Soueidan & Nikolski, 2016)	Open source
	Deep learning	Meta-MFDL (Yang, et al., 2021)	Open source
		HyperVR (Ji, et al., 2023)	Open source
		Orphelia (Soueidan & Nikolski, 2016)	Open source
		CNN-MGP (Al-Ajlan & El Allalli, 2018)	Open source
		PHANOTATE (McNair, Zhou, Dinsdale, Souza, & Edwards, 2019)	Open source
Functional annotation	Deep learning	MetaMLP (Arango-Argoty, Heath, Pruden, Vikesland, & Zhang, 2019)	Open source
		DeepFRI (Gligorijević, et al., 2021)	Open source
	Naive Bayes Classifier	RDP classifier (Krause, et al., 2021b)	Open source
Taxonomic annotation	Deep learning	GeNet (Rojas-Carulla, et al., 2019)	Open source
		BERTax (Mock, Kretschmr, Kriese, & al, 2021)	Open source
		DeepMicrobes (Park, et al., 2023)	Open source
Feature selection and extraction	Decision tree	Xtreme Gradient Boosting (XGBoost) (Krause, et al., 2021b; Jiang, et al., 2023)	Open source

Phylogenetic analysis	Machine Learning	UShER (Turakhia, Thornlow, & Hinrichs, 2021)	Open source
Phenotype classification	Support Vector Machines	VirulentPred (Ji, et al., 2023)	Open source
	Deep learning	DeePaC (Bartoszewicz, Seidel, Rentzsch, & Renard, 2020)	Open Source
		DeepARG (Arango-Argoty, Heath, Pruden, Vikesland, & Zhang, 2019)	Open source
		DeepTE (Yan, Bombarely, & Li, 2020)	Open source
		DCiPatho (Jiang, et al., 2023)	Open source
		BacPaCS (Barash, Sal-Man, & Sabato, 2019)	Open source
		DeepForest (Zhou & Feng, 2017)	Open source

***Source**: author's elaboration based on literature review findings.*

*[a] For this study, considering that the vast majority of tools is used for research purposes and in tight collaboration between the industry, academia and the clinical settings, the availability of the tool refers to the type of access to the source of the tool, distinguishing between "**open source**" and "**proprietary**", indicating the owner of the tool where possible*

2.1.3 AI tools for metagenomics data summary and visualisation

These are the main takeaways on the role of AI-based tools enhancing the summary and visualisation of metagenomic sequences.

- There is **scarcity of AI-based tools specifically designed for data summary and visualisation in metagenomic studies.**
- Both the literature and the experts point to the **need for innovative solutions in this area**. However, it is noted that **the way forward to innovation doesn't necessarily go through the inclusion of AI algorithms**. The interactivity of the tools and a smooth user experience play a major role. **Bioinformatic pipelines for metagenomic analysis, some of them integrating AI algorithms, can facilitate and standardise the workflows in metagenomic studies**. AI-based tools show potential in the summary of complex data like large longitudinal surveillance studies or in cases involving multidimensional variables. **AI-based tools could bridge the gap integrating different datasets with various data types.**

The third and final phase of a metagenomic study involves the **preparation of interactive data summaries and visualisations to explore and interpret the data generated at the various antecedent steps.** This phase includes steps like building correlation networks and composition diagrams, and clustering visualisation.

Metagenomics analyses result in large amounts of complex data, often difficult to analyse and interpret. Effective data summary and visualisation become an indispensable part of exploratory data analysis in metagenomics (Sudarikov, Tyakht, & Alexeev, 2017). However, the heterogeneity of the sample data sources, sequencing approaches, data formats and outcomes of metagenomic analyses make robust data visualisation difficult.

The literature points to the importance of tools and resources for data summary and visualisation in metagenomics research. Some authors support that it is essential to develop interactive tools to allow unspecialised healthcare professionals to grasp a better understanding and interpretation of the metagenomics data (Song, Su, Xu, & Ning, 2012). In

fact, Reis et al. (2021) present a conceptual architecture for AI-based Big Data Analysis and Visualisation supporting metagenomics research based on the AI2VIS4BigData Reference Model. The authors claim that by combining computational power with visualisation capabilities, the proposed framework could enhance researchers' understanding of microbial communities and their interactions, ultimately advancing the field of metagenomics research.

Although scarce, **the literature search has identified some cases where AI tools are integrated into the data processing of visualisation tools**. An example is the open-source tool "Interactive Contig-bin Verification and Refinement tool" (**ICoVeR**) (Broeksema, et al., 2017) that, based on multiple binning algorithms, allows the visualisation of the generated genomic bins in different modes, including a dimensionality reduction plot, facilitating the curation of the bins. In fact, as reported in Aplakidou et al. (2024), several tools integrate dimensionality reduction algorithms to facilitate the visualisation of complex and high-dimensional data. Some of the most used methods include Uniform Manifold Approximation and Projection (UMAP), t-distributed Stochastic Neighbour Embedding (t-SNE) and the probabilistic generative model Latent Dirichlet Allocation (LDA). Another notable approach is **VizBin** (Laczny, et al., 2015) , which uses self-organising maps (SOM) for dimensionality reduction. This method enables reference-independent visualisation and human-augmented binning of metagenomic data by clustering contigs into bins. Emerging self-organising maps (ESOM) are unsupervised neural network algorithms that represent multidimensional data in two-dimensional clusters. Laczny et al. (2014) also reported a **scalable approach for the visualisation of metagenomic data** based on dimension reduction via Barnes-Hut Stochastic Neighbour Embedding (BH-tSNE). This approach allows an alignment-free assessment of the taxonomic structure of the genomic fragments analysed. Visualizing the diverse communities in a metagenomic sample is challenging when there is no prior taxonomic information. Kouchaki et al. (2016) presented an alternative **metagenomic visualisation approach** that combined local binary patterns (LBP), randomised singular value decomposition (RSVD) and BH-tSNE that reflected the underlying taxonomic structure of the metagenomic data.

However, these visualisation techniques have an important limitation in metagenomic surveillance, where the amount of data makes it unlikely to fit into a screen visualisation. There are several tools developed to facilitate and enhance the interpretation of AI-based tree ensemble models like Random Forest. endoR is an example which is used to facilitate the interpretation of phenotype prediction by simplifying the fitted model into a decision ensemble and extracting information on the importance of individual features and their pairwise interactions, reducing the complexity and ensuring that only essential parts of the model are retained (Ruaud, Pfister, Ley, & Youngblut, 2022). However, these tools are not necessarily based on AI.

The results of the desk research show that there are multiple and diverse tools to facilitate the interpretation of the result of metagenomic analyses (see Aplakidou, et al., (2024) and Sudarikov et al., (2017)). However, **most of these tools do not rely on AI. Various authors explain this by the accuracy and efficacy of standard methods for completing visualisation tasks** (Krause, et al., 2021b).

***Box 5**. Considerations from the experts: AI tools for metagenomics' data summary and visualisation*

Highlights resulting from Experts' consultation

All the experts interviewed agreed that the **current landscape of AI tools specifically designed for data summary and visualisation in metagenomic studies is relatively sparse**. Traditional metagenomic tools primarily focus on the presence/absence or quantification of specific organisms. However, visualising this complex data effectively requires innovative approaches and substantial computational resources.

Many experts agree that AI has the potential to significantly enhance data summary and interpretation in metagenomic studies. An expert from the industry sector emphasised

the benefits of **using AI-based tools for trend analysis in longitudinal surveillance studies**. AI could facilitate the comparison of multiple datasets over time, revealing trends that might otherwise be overlooked. Other experts with experience in public health surveillance campaigns noted that **AI could be especially beneficial in cases involving multidimensional variables, such as in foodborne outbreak investigations**, as AI-based tools can identify patterns that traditional tools might miss, offering a more nuanced understanding of the data.

Some experts, when considering the potential use of AI for building correlation networks, noted that there are numerous well-developed tools for epidemiologic data. However, applying these tools to genomic data is challenging due to the lack of data integration. These experts shared that this could present an opportunity for **AI to bridge the gap integrating different datasets with various data types** to provide comprehensive insights.

Another aspect discussed is the **relevance of bioinformatic pipelines for metagenomic analysis that can facilitate and standardise the workflows in metagenomic studies**. There are numerous pipelines designed for analysing Nanopore sequence data, and many include tools like PANGOLIN, which includes pangoLEARN, an algorithm trained using SARS-CoV-2 sequences that is currently using Random Forest and is used for lineage assignment for lineage assignment. Additionally, bioinformatics pipelines such as Pathogenwatch, Nextstrain, and Microreact integrate data visualisation, making the data interactive and accessible. Despite these advancements, many pipelines have yet to incorporate AI.

The analysis of the literature has shown that the phases of the metagenomic workflow where a greater variety of AI techniques are employed are the **Annotation and Analysis** steps. More specifically, many of the identified AI-based tools are focused on enhancing gene prediction and functional annotation of metagenomic samples. The most used machine learning technique across the different phases of metagenomic studies is deep learning due to its ability to learn intricate patterns from complex and high-dimensional data.

The next section will review the primary bottlenecks identified in existing literature concerning the use of metagenomic analysis and whether AI-based tools can help to overcome them.

2.2 Identification of current bottlenecks and need of innovation in metagenomics for pathogen identification

In the past decade, the field of metagenomics has experienced great progress and developments, allowing a better understanding of complex microbial communities. Nonetheless, several bottlenecks persist, and further innovation is needed to overcome these challenges and drive the field forward. The review of the scientific literature provided a general overview of the bottlenecks identified in the implementation of metagenomic analyses in clinical or environmental monitoring. The main bottlenecks identified in metagenomics for pathogen identification and health preparedness are:

A. Availability and accuracy of the data.
B. Complexity of the data.
C. Availability of reference data.
D. Increased requirements of computational resources.
E. Genomic data storage and transfer requirements.
F. Need for upskilling and capacity building among professionals.

Below, the most pressing bottlenecks identified in the literature are described together with potential solutions posed using AI.

A. Availability and accuracy of the data

Metagenomic analyses are highly dependent on the **quality of the data analysed**. Undersampling, sample contamination, storage bias, low accuracy in base calling or assembly errors are among the initial challenges that may jeopardise the accuracy and reproducibility of metagenomic studies (Kumar, Lorusso, Fosso, & Pesole, 2024) (Duarte, et al., 2020) (Lema, Gemeda, & Woldesemayat, 2023). Moreover, it is essential to consider the **high dimensionality of metagenomic data**, as the large number of features associated with the microbial community samples can make data analysis computationally challenging and prone to overfitting.

As Pödör & Hekfusz (2024) highlighted, *"data preprocessing is the foundation of successful and reliable data analysis",* thus appropriate preprocessing steps, such as data quality assessment or feature selection, are essential to ensure the quality and reliability of subsequent analyses. The combination of errors and biases in sequencing technologies, incomplete databases, similarity among otherwise divergent genomes, and the large volumes of sequencing data required for metagenome sequencing has led to unacceptably high false discovery rates (Freitas, Li, Scholz, & Chain, 2015). In 2016, a diverse set of stakeholders came together to provide a guiding framework for the accessibility and reuse of scholarly data, known as the FAIR Guiding Principles (Wilkinson, Dumontier, Aalbersberg, & al., 2016). Now widely adopted as standards for data management, the principles aim at providing data that is Findable, Accessible, Interoperable and Reusable. As previously reported, AI techniques offer promising solutions to address these challenges. As observed in the literature reviewed, **SVMs have been used to optimise the base-calling of metagenomic sequencing**. **ML algorithms can be trained on large datasets of known microbial genomes to learn patterns distinguishing different taxa**. These models can then be used to classify metagenomic reads more accurately than traditional methods. **AI-based techniques can also be used to improve the accuracy of annotations,** and techniques such as **autoencoders or PCA can be used to reduce the dimensionality of metagenomic data**, making it more manageable for analysis and interpretation (Pödör & Hekfusz, 2024).

Moreover, it is essential to consider that metagenomic datasets are compositional, as they don't contain absolute values on the organisms present in a sample but information on the relationships between the parts' relative abundance measures (Gloor, Macklaim, Pawlowsky-Glahn, & Egozcue, 2017). Therefore, there isae need to **employ compositional data methods when applying AI to metagenomic datasets to avoid misleading and unpredictable results** (Gloor, Macklaim, Pawlowsky-Glahn, & Egozcue, 2017). Recent research identifies a generally limited adoption of transformation methods targeting the compositional nature of this type of data. Many studies tend to use relative and normalization-based transformations that could lead to less accurate analyses (Ibrahimi, et al., 2023).

B. Complexity of the data

Metagenomic data is complex, not only because of the high number of genomes and features present in a sample but also because of the complex and non-linear interactions within the microbial community represented. Classical statistical methods used in metagenomics tend to rely on a priori knowledge and only assume linear interactions, thus not capturing the complex nature of the sample under study (Wijaya, Park, Yang, Siddiqui, & Oh, 2024). **ML modelling can follow a research strategy not based on a priori knowledge able to capture intricate non-linear relationships in metagenomic samples.** It is therefore considered that AI techniques improve the performance of the analysis of samples coming from complex

ecosystems like the soil, water and human body (Ghannam & Techtmann, 2021). It is also important to consider that besides the complexity of the metagenomic data itself, repositories of metagenomic data should include metadata, the contextual information of the sampling and experimental conditions as well as the preprocessing and feature extraction steps (Kumar, Lorusso, Fosso, & Pesole, 2024). **Metadata, which is essential for conducting robust data analysis, is very rarely available in metagenomic repositories**. Potential solutions to this bottleneck have been offered by a tool developed by microbiome epidemiology researchers, the checklist known as the STORMS guidelines (Mirzayi, et al., 2021) that offers guidance for concise and complete reporting of microbiome studies and the FAIR principles for data management (Wilkinson, Dumontier, Aalbersberg, & al., 2016). Moreover, ML also offers the possibility of increasing the quality of databases' metadata by automating the curation process. However, the accuracy of these algorithms is still not optimal**. Further research is needed to expand the sources for metadata extraction and the development of metadata standards** (Kumar, Lorusso, Fosso, & Pesole, 2024).

C. Availability of reference data

Another important challenge in metagenomic analyses is the **limited availability and bias of reference databases**. The partial representativity of these databases impedes the accurate identification and classification of microbial sequences, particularly for novel or uncultured microorganisms (Lema, Gemeda, & Woldesemayat, 2023).

Advancements in high-throughput sequencing technologies have resulted in the rapid accumulation of genomic data from microbial communities. However, reference genome databases lack representation for the vast diversity of microbial life, and most microbial gene functions remain uncharacterised (Maranga, et al., 2023; Quince, Walker, Simpson, Loman, & Segata, 2017) The billions of still unknown novel species and genes and dynamic patterns within the microbial communities form the so-called microbial 'dark matter' (Zha, Chong, Yang, & Ning, 2022). Authors (2022) found that AI technologies (for example, DeepARG (Arango-Argoty, et al., 2018)) can be leveraged to enable more efficient mining of microbial dark matter.

Moreover, **AI can contribute to the expansion of the current genomic databases through its demonstrated potential to increase the accuracy and processing times of reference-independent metagenomic methods** (de novo assembly). Large de novo assembly could contribute to and resolve ambiguous regions of existing reference genomes, as well as providing additional data for the generation of reference genomes of previously uncharacterized species. By expanding the representativity of the current reference databases with labelled genomes, AI can contribute to faster and simpler identification of pathogens through metagenomic analyses.

D. Increased requirements of computational resources

As previously mentioned, the study of metagenomics has greatly benefited from low-cost and high-throughput sequencing technologies (e.g., MinION, Illumina Sequencing technologies), which have led to fast and inexpensive production of large amounts of biological sequence data (Li, et al., 2016). However, the **fast-increasing volume of sequencing data poses an important challenge to computational methods, which now require a substantial increase in computational resources.**

The NGS assembler MEGAHIT (Li, et al., 2016) was improved by using ML techniques that were built using specifically CPU-based algorithms, eliminating GPU dependencies and the need for specialised hardware on the user's end (AWS), as well as overcoming the memory bandwidth bottleneck, a common problem of GPUs in bioinformatics tasks like genome assembly.

Nanopore sequencing technology offers real-time and in-field analysis of the samples, allowing the sequencing of samples directly at the point of care (Wang, Zhao, Bollas, Wang, & Au, 2021). However, the resulting sequences require additional processing. Chiron, NanoDeep and FastBonito translate nanopore raw signal directly into nucleotide sequencing, these **Deep learning models improve the base-calling accuracy without losing speed and without specialised computer resources, resulting in a reduction of computational resources**.

E. Genomic data storage and transfer requirements

As reported by Lema et al. (2023), **the vast amounts of data generated by next generation sequencing and metagenomic studies pose significant challenges in terms of storage and transfer, creating a bottleneck in the analysis pipeline**. As metagenomic datasets often consist of billions of short DNA reads, storage requirements can quickly escalate, necessitating large-scale data centres or cloud-based solutions. Moreover, the transfer of such massive datasets between sequencing facilities, data centres, and research labs can be time-consuming and expensive, hindering collaboration and delaying analysis.

AI can be leveraged to reduce the effect of these key challenges in metagenomics analysis. First, as presented before, **AI-powered data preprocessing techniques can filter out low-quality or redundant data, reduce dimensionality and extract key features, reducing storage and transfer requirements**. However, this promise of AI needs to be taken with precaution as tools need to be thoroughly evaluated to ensure the proper extraction of features that would allow the detection of novel pathogens without losing key information in the processing of the data. Second, studies (Zhang, Zhang, & Xu, 2022) showed that AI can be used to minimise bandwidth usage and accelerate analysis. This could be leveraged for metagenomics studies as well. In the age of information, the dilemma of data storage and transferability extends beyond metagenomic analysis. Therefore, several initiatives are actively exploring how innovative solutions, like AI, can serve as a key asset to mitigate this challenge. A selection of innovative solutions for data storage and transfer tailored to the field of metagenomics is presented here. At an individualised or localised scale, Manter et al. (2016) proposed a local web server for the storage and analysis of metagenomic data, myPhyloDB**.** This personal database has **a browser interface designed to facilitate the individual storage, processing, analysis, and distribution of microbial community populations**. It also integrates some bioinformatic tools for processing raw sequencing data like Mothur (presented above) and performing analyses. This tool simplifies the processing of data by providing standardised metadata, a set of normalisation techniques, and allows the customization of analytical pipelines. The authors emphasise that despite its main use as a personal application, myPhyloDB could be used as a basic component of a further developed institutional web-server for metagenomics data analysis and distribution platform (Manter, Matthew, & Delgado, 2016). Other initiatives have focused on **developing secure cloud-edge computing architectures for metagenomic analyses with a special focus on supporting the use of portable sequencing machines in the field** (D'Agostino, Morganti, Corni, Cesini, & Merelli, 2019). The increasing popularity of portable sequencing machines has resulted in the generation of a collective network of devices connected to a computing application to get insights from the device-generated data. This generates huge amounts of data that are difficult to manage, considering the number of resources that proper data transfer and processing require. The current solution for data transfer between physical devices (portable sequencing machines) and the computing application consists of sending the data to a central Cloud node for analysis (Verderame, et al., 2020). However, with the expansion of users, the Cloud node will become a bottleneck with an overloaded bandwidth, and the current solution will be inoperable. Verderame and colleagues (2020) propose the use of the Cloud-Edge computing paradigm to split the analysis workflow between the data source and the Cloud infrastructure, preprocessing the raw sequencing data to reduce the volume of data transferred to the Cloud infrastructure that can further process and store the data. Recent work has also focused on

the **design of a computational server system aimed at processing and analysing bioinformatics data in metagenomics using the specialised software product Galaxy Europe** (Kirilov & Kirov, 2023). Galaxy Europe is a software platform accessible through a web browser that connects the user to a server, enabling collaborative research. As reported by Kirilov and Kirov (2023), Galaxy Europe is an easy solution for non-expert programmers that supports various types of data and integrates many tools used for data analysis like the Mothur tool presented above for metagenomic OTU clustering. Integrating a server system that can run parallel computations can significantly increase the speed of data processing.

F. Need for upskilling and capacity building among professionals

Advances in sequencing technologies have enabled culture-independent profiling of microbial community membership and function. These advancements have rapidly expanded the knowledge of human-associated and environmental microbiomes, which makes metagenomics a key asset in the context of human health (Marić & Šikić, 2019). At the same time, **advances in the field have produced massive amounts of data, which requires skilled researchers to analyse and interpret it**. Metagenomic data analysis requires expertise in bioinformatics, computational biology, AI, and data science due to the complexity of the data and the computational methods involved. In fact, experts in the field have stated that "there is no best use scenario when it comes to ML, and that individual researchers should select methodologies that are consistent with the specific domain problem, the questions being asked, and based on available data" (Ghannam & Techtmann, 2021). However, researchers and clinicians often encounter challenges in selecting appropriate analytical tools, troubleshooting computational pipelines, and interpreting the results (Lema, Gemeda, & Woldesemayat, 2023). Therefore, there is a growing need for upskilling to ensure that professionals from multiple disciplines that work with metagenomic data (clinicians, bioinformaticians, geneticists, data scientists, etc.) possess the necessary skills to effectively analyse and interpret metagenomic data.

Some efforts have been made to create **microbiome-specific bioinformatic pipelines that integrate user-friendly ML implementation that can be accessed through web-interfaces and common programming languages like R and Python** (Ghannam & Techtmann, 2021). These metagenomic pipelines streamline the process by offering a step-by-step approach to metagenomic analysis. Several specialised tools are integrated into these pipelines where the output of one tool can be directly used by another tool. This guided approach to metagenomic analysis facilitates the work of the researcher, who can reuse pipelines for similar studies and adapt them to their particular needs, if required. Besides the already mentioned Galaxy Europe (The Galaxy Community, 2022), there are other well-known bioinformatic pipelines that can be used for metagenomic analyses like MGnify (Richardson, et al., 2023) and QIIME 2 (Bolyen, et al., 2019).

***Box 6**. Considerations from the experts: identification of current bottlenecks and need for innovation*

Highlights resulting from Experts' consultation

The experts' consultation validated the bottlenecks already identified in the literature and presented above, with some notable comments both on the bottlenecks and need of innovation.

All stakeholders agreed that the bottlenecks that stand out the most are those concerning the **volume, complexity and accessibility of the data and regarding the need for computational power to get through such large volumes of data**. It is essential to highlight that different stakeholders (from each category) also pointed out that the **privacy of the patients must be protected** (also when dealing with metadata), and, in this regard, the **EU** could do better both in terms of **providing clearer guidelines on**

how to handle patients' data and in providing the infrastructure needed so that researchers would not have to resort to Big Tech US players like Google. In fact, uploading personal data on tools and servers owned by non-EU companies poses an issue for data privacy and **transparency**, as insight on how the tool functions are not provided. The **EU** could also help in enhancing data interoperability among countries, by creating standardised protocols.

Standardisation of protocols and accessibility are other relevant issues among clinicians and the industry: datasets from different samples of research groups (that often use different protocols) need to be more compatible as standardised wet lab procedures would significantly reduce variability and enhance the ability to compare data across studies. Moreover, a lot of data is being processed in several research institutions and it would be good if the access to this data could be shared with other researchers, as the datasets from clinical practice tend to be small, bigger datasets would be more informative. Accessibility to **metadata** is also a relevant issue.

All stakeholders agreed that there is a **need for upskilling** among professionals and one stakeholder from the industry pointed out that the **need for upskilling is slowly being addressed** by the industry (Oxford Nanopore being one notable example) which is developing simpler solutions (fully automated instruments with integrated sequencers) to be used by non-specialists.

One bottleneck that was not encountered in the review of the literature but mentioned by stakeholders is **the need to integrate genomics and epidemiological data which are currently compared but not integrated**. In this regard, AI tools could be of help.

One stakeholder from the academia pointed out that **coverage estimation** is another big challenge in their daily work: understanding the representativeness of our sample is crucial and currently, there are no instruments to **assess the coverage or amount of sampling required for it to be representative of the ecosystem under study**.

A couple of stakeholders from different categories (government and industry) both pointed out the challenge of accessing, transferring and using **large amounts of sensitive data for research purposes** and both mentioned a solution involving secure data storage and controlled access for researchers. The industry stakeholder further suggested that federated networks could behave as a middleman to facilitate research while protecting data.

2.3 Additional inputs shared by experts on the priorities and funding opportunities of the EC, the collaboration between the industry and researchers, and some additional considerations on the use of AI

This section of the report summarises the experts' contributions worth noting and not directly covered by the research questions that guided the findings presented above.

***Box 7**. Considerations from the experts: Additional inputs*

Highlights resulting from Experts' consultation

1. On the perceived priorities of the European Commission:

According to the experts' consultations, DG CONNECT has focused on three initiatives to deliver **key data infrastructure** in the healthcare sector: the Genomics initiative, the

European Cancer imaging initiative, and the Virtual Human Twins initiative. These initiatives were launched with specific policy goals such as enhancing personalised medicine and improving the efficiency of health healthcare systems. Specifically, according to stakeholders, AI has gained prominence across the three mentioned initiatives.

Given that metagenomics is a relatively new field it has yet to be prioritised by DG CONNECT. Currently, metagenomics is not a key funding priority for the DG CONNECT, DG RTD, and DG SANTE agencies, according to an expert from one of these agencies.

However, during the interview it was noted that a recent funding opportunity under the EU4Health program by HaDEA and DG SANTE focuses on the use of AI for metagenomics. One academia stakeholder states that metagenomics has been the focus of various exiting EU initiatives (e.g., HERA: “Genomic surveillance initiative”, EU-WISH: “wastewater surveillance initiative”, EU-HIP: “initiative for enhanced IT systems and interoperability”, DURABLE: “preparedness initiative”). Particularly, one existing funding initiative aimed at improving pathogen detection and surveillance was EU4Health, "Point of care metagenomic sequencing for universal pathogen detection" (2023).

The experts’ consultations described what they perceive as the main EU priorities (i.e., basic bioinformatic workflows, data management, data repositories, data integration; metagenomic data analysis; and metagenomic surveillance). **Basic bioinformatic workflows** include tasks such as genome assembly and the detection of pathogenicity and virulence factors. Currently, the priorities are to ensure these methods are reliable and accessible to non-bioinformatics experts. Additionally, it is important to make these methods scalable and robust. The development of guidelines and thresholds, such as those used to define an outbreak is often crucial.

According to stakeholders, the following should be prioritised: improved **data management, data repositories, and data integration systems** that can bring together various types of data, including metagenomic, epidemiology, and from genome wide association studies (GWAS). **Metagenomic surveillance** is a promising approach that should be considered as supplementary to the existing clinical surveillance. However, the analytics process in this field is still under development, and more efforts are required to interpret this data and better understand its implications for public health measures.

2. **On EU funding:**

With regards to funding related to clinical metagenomics, stakeholders shared that it should be allocated into three main contexts: targeted metagenomics for fast diagnostic on life threatening infections; clinical metagenomics for chronic infections and pathogen discovery. Currently, multiplex testing is the standard in clinical settings, but targeted metagenomics represents a promising innovative initiative for rapid diagnostics. In cases of chronic infections, identifying the causative microorganism can be challenging because it may have been killed by antibiotics. Additionally, metagenomics holds potential for pathogen discovery, expanding the ability to detect and to better understand pathogens.

According to another stakeholder from public health laboratories, EU funding should be allocated for **AI and adequate infrastructure** because many large AI models are hosted on U.S. servers, which makes it difficult given that the EU data regulation policies do not apply.

Stakeholders from the academia and clinical laboratories feel that at the European level, having resources to host AI models and data would be advantageous. For example,

Google, Microsoft, and OpenAI, all provide substantial research space, but Europe lacks comparable facilities. Google, for instance, offers server access and model time for research, which is not as costly given that the infrastructure already exists. Furthermore, it is not easy to access EU funding / research opportunities as they are not as advertised as Google / Microsoft's funding. Furthermore, clear guidelines on using AI in medical and surveillance contexts would be helpful because establishing rules on data usage and permissions can clarify the framework for researchers, which can benefit the scientific community.

According to government stakeholders, EU funding is very broad and has a wide selection of topics Several EU funded projects are focused on AI-based tools, both on developing the tools and evidencing the value of using these tools.

However, despite the EU initiatives mentioned by the academia stakeholder listed above, most government stakeholders argue that it would be of added value if funding was distributed evenly across other related disciplines that are equally important for pathogen identification and outbreak control. Moreover, funds could be employed for the digitalisation of the health sector. For example, improving the standardisation of the EHR would allow for better gathering of additional data (i.e., clinical data) that could accompany the genomic data. Which can be beneficial for public health and research sectors (e.g., in this regard, it is difficult to have a harmonised system in the EU).

3. **On industry and academia collaboration:**

The perfect combination of three parties (i.e., academic, industry, and implementers such as clinicians, researchers, or private companies) can drive innovation and the application of metagenomics. The academic sector develops new technologies, industry focuses on scaling up on innovation, and implementers apply tools in a practical setting. Big consortium grants are instrumental in bringing diverse stakeholders together. However, a significant bottleneck in this aspect prevents the industry from scaling up tools unless there is a great market opportunity. Thus, EU support can bridge this gap by facilitating large consortiums that unite key players to refine and tailor the right tools for specific market needs.

According to stakeholders, many new industry initiatives or collaborations aimed at leveraging AI-driven approaches to optimise metagenomic analysis for healthcare applications began in the aftermath of COVID-19. Some examples are the **ARTIC network** which, through the collaboration of academics, compiled a wide range of data that generated large databases of sequences. Consequently, there are different players starting to work in this specific area such as the **nucleic acid observatory**. Stakeholders expressed their support to further explore multidisciplinary collaboration opportunities driven by HERA, industry, academia, and other relevant stakeholders. **Additionally, collaboration and cooperation between EU member states** also plays a vital role in these efforts and could be leveraged at the EU level.

4. **Key considerations for the use of AI:**

Stakeholders emphasised that there must be an economic analysis for the implementation of these AI-based tools in metagenomics. However, first, exploring **anonymization is crucial because it involves removing** identifiable human information from data to protect individual privacy and confidentiality, which can address concerns about transparency. For instance, regarding the adoption of AI technology, an interviewee recalled the importance of building **trust** among clinicians and the healthcare system in general, and this trust includes the security of the data being used. Additionally, one Clinician shared valuable insight on how their unit overcame the issue of patients'

anonymization, which was by substituting their data with the accessible data catalogued in the 1000 human genome project (https://www.internationalgenome.org/).

Moreover, it is important to expand on the explainability of AI tools because they often lack confidence intervals, which is important for their explainability. Lastly, with regards to promising innovations, **Natural Language models** hold for the processing of currently available data and how it could **inform actions in public health.**

At the moment, AI tools for metagenomics are probably 90% open-source, instead the infrastructure is proprietary. One notable example is GINKGO−bioworks, an open community that relies on open-source (https://www.ginkgobioworks.com/).

3. Discussion & Conclusion

In this study, we aimed to **explore the use of AI-based tools in metagenomic analysis in the context of pathogen identification for health emergency preparedness and response**. Through an extensive review of the literature and consultation activities with experts in the field, we investigated the current availability and use of these tools, as well as the main bottlenecks existing in the field of metagenomics and the adoption of AI-based tools. This section will dive into the potential impact of AI-based tools for metagenomic studies on health emergency preparedness and response and propose some actions that could enhance the proper adoption and implementation of these innovative solutions in the field.

When considering metagenomic studies, it is essential to consider that metagenomics can both be used for clinical diagnostics (clinical metagenomics) and for systematic surveillance on both the environment (e.g., wastewater, air ducts in hospitals, etc.) or asymptomatic population. Due to the different challenges posed by the application of metagenomics in these scenarios, it became clear from the consultation with experts that it's crucial to understand the origin of the sample and the expected output that metagenomic analysis can deliver in order to properly identify the type of analyses that need to be conducted and the phases that can be optimised by the use of AI.

This study has identified a series of bottlenecks for the proper adoption of metagenomics for both clinical diagnostics and surveillance. The main bottlenecks identified initially in the literature have been the availability and accuracy of the data, the complexity of the data, the availability of reference databases, the increased requirements of computational resources, the requirements of genomic data storage and transfer, and the need for upskilling and capacity building among professionals. The consultation with experts in the field provided additional insights, both validating the pre-identified bottlenecks and recalling some additional ones, i.e. the accessibility of the data across research or governmental institutions, the standardisation of the protocols, and the integration of the genomic data with epidemiological data.

In this report, existing tools and AI techniques were mapped according to their use in the different steps of the metagenomic workflow, grouped into three phases for the purpose of this study (data preparation and aggregation; annotation and analysis; and summary and visualisation). The results show the clear role of Deep Learning techniques in increasing the accuracy of base calling, especially for those novel sequencing technologies that are faster and allow for onsite direct analysis. Clustering and dimensionality reduction AI algorithms, despite being simpler machine learning algorithms, play a crucial role in the aggregation of metagenomic sequencing data. Despite the opportunities that AI-based tools present for metagenomic data preparation and preprocessing, the literature and consultations with experts points to a **major contribution of AI in the metagenomic data annotation and analysis phase**. Both the literature review and expert consultations indicated consensus that deep

learning techniques show the potential to improve the accuracy and speed of gene prediction and functional annotation, especially for novel sequences with limited similarity to reference databases. These improvements are facilitated by automating traditionally time-consuming steps in the metagenomic workflow and by accommodating a diverse range of tasks and processes. This becomes extremely relevant to disentangle the metagenomic “dark matter” and the limited availability of reference databases for most of the microbial realm, which was often highlighted in the literature review. Finally, regarding the use of AI-based tools for metagenomic data summary and visualisation, the contributions of AI to this area are more limited. However, consultation with experts showed a general interest in the development of interactive tools that enable researchers to engage with their data and outcomes. In this regard, it is noted that the relevance of bioinformatic pipelines is to integrate complex protocols and present the results of the data analysis to the user. As observed in the literature and learned from experts in the field, current developments in bioinformatic tools for metagenomic analysis are partially addressing the bottlenecks mentioned above. For example, the improvement in base calling efficiency and the rapid recognition of complex patterns improve the processing times and the need for expert manual curation of raw data. Moreover, AI-based tools perform better in reducing the complexity and dimensionality of the data, reducing the requirements of data storage and computational resources and facilitating data transfer and accessibility.

Overall, the **adoption of AI-based tools in metagenomics can enhance the EU's crisis preparedness and response capabilities by enabling a more rapid, accurate and efficient pathogen identification compared to traditional tools**. These tools can facilitate the early detection of emerging pathogens, ensuring a timely intervention and effective allocation of resources during public health crises. Moreover, integrating AI techniques into existing surveillance and monitoring systems can strengthen the EU's ability to anticipate and mitigate the impact of infectious disease outbreaks, safeguarding public health and protecting against future health threats. AI-enhanced metagenomics not only facilitates pathogen identification but also contributes to the detection of antimicrobial resistance genes in those pathogens and could improve the diagnostic process, from disease classification to prediction of clinical outcomes (Peiffer-Smadja, et al., 2020).

However, the **adoption of AI technologies in metagenomics calls for strategic policy actions to maximise their potential benefits**. From the information gathered by reviewing the literature and consulting experts with various backgrounds in the field, it became clear that there are a series of policy actions that could enhance the use of metagenomics for health preparedness and response using AI-based tools.

1. **Provide spaces and resources to strengthen the collaboration among diverse stakeholders in this multidisciplinary field**. All the experts interviewed emphasised the critical need to create more collaboration opportunities, enabling the proper assessment of the field’s needs and accelerating the development of solutions. Key suggestions included organising multidisciplinary conferences that bring together experts from diverse backgrounds like computing sciences, biological sciences and clinical practice. These spaces would foster dialogue and inspire future collaborations. Additionally, financial support for research initiatives, especially public-private partnership initiatives, can facilitate the engagement between the industry and academia, streamlining the development of AI tools and their launch to the market. Moreover, this action should also consider the promotion of capacity-building initiatives among researchers and professionals working on public health, for example, training workshops on AI implementation and governance.
2. **Support the validation and evaluation of existing tools**. The use of AI-based tools depends largely on the trust of the user. Therefore, to increase the adoption and impact of existing AI tools, it is essential to establish processes for their validation that could provide evidence to increase the trust of potential adopters. In this regard, the standardisation of validation frameworks to ensure consistency and reproducibility, the provision of funding for collaborative validation of AI tools and the establishment of

regulatory guidelines for the validation of AI-driven metagenomic tools could be specific actions that could support the adoption of AI technologies in metagenomics. Properly evaluated tools, backed by transparent and documented results, will have a faster and easier implementation roadmap.

3. **Provide the resources to facilitate data sharing across Member States and research institutions**. A key factor in the development of AI tools for metagenomic analysis is access to large datasets. A good data sharing framework to facilitate the accessibility to the data while ensuring the protection of personal data is essential. More specifically, the adoption of unified criteria for data and metadata sharing is pivotal in fostering the effective application of AI and ML methods within the European Union. By promoting transparency and facilitating data exchange among various stakeholders, these policies create a more conducive environment for AI and ML development. For example, the FAIR principles, which advocate for data to be Findable, Accessible, Interoperable, and Reusable, serve as a valuable framework for guiding data-sharing practices. Additionally, the GDPR and the European Health Data Space (EHDS) provide essential guiding frameworks. However, the implementation of these frameworks – particularly the EHDS – remains inconsistent across Member States, sometimes leaving stakeholders uncertain about how to proceed. Moreover, open data practices could be incentivised by the requirement for EU-funded projects to contribute datasets to shared repositories (if ethically and legally possible) and adopt open data-sharing practices. As reflected by several experts, the EU could introduce metadata standards for metagenomic datasets to increase compatibility and facilitate data sharing.
4. **Provide additional support to continue developing AI-based solutions tailored to environmental metagenomics** to contribute to the detection and prevention of major outbreaks in a timely and efficient manner by improving systematic surveillance methodologies. Compared to clinical metagenomics, environmental metagenomics deals with much bigger and complex datasets where AI could play a bigger role. Therefore, dedicated funding programmes for the development of AI-based technologies applied to environmental metagenomics could ensure long-term investment in developing tailored solutions in the field. Moreover, monitoring the development of AI technologies in environmental metagenomics could contribute to assessing the progress and impact of these new technologies.
5. **Establishment and correct implementation of regulatory frameworks and standardised guidelines to ensure the proper use of AI**. These guidelines could include specific requirements for transparency, accountability, and fairness in AI systems, as well as protocols for ensuring data privacy and security.

These are some of the key policy domains that must advance hand in hand with the development of AI and metagenomics technologies to ensure the advancement of the EU in health emergency preparedness and response. Further lines of research following this report could be to investigate further the role intermediate hosts play in transmitting and harbouring microorganisms that impact human health. This is particularly relevant in the case of zoonotic diseases, understanding the role, for example, of birds as intermediate hosts for certain viruses, can help guide public health measures.

By addressing the existing challenges and investing in AI research and development, the European Union can harness the full potential of metagenomics and AI to strengthen its health emergency preparedness and response capabilities. This will ultimately contribute to a safer and healthier future for all European citizens.

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Annex 1: Methodology

This section outlines the methodology followed in this report for the review of the literature on the use of AI-based tools for improving novel pathogen identification through metagenomic analysis. Moreover, this section also describes the methodology used to gather additional inputs on the topic from key stakeholders, covering from the selection of stakeholders up to the analysis of the inputs gathered.

Scientific literature review

The focus of the systematic literature review was to analyse the literature reporting on how further innovation in software aspects, especially AI-based tools, can improve the identification of pathogens through the analysis of metagenomic data retrieved from different samples (both clinical, environmental and human microbiome) and how novel tools can help reduce genomic data storage and transfer requirements. An extensive literature search was conducted following the PRISMA guidelines and using the academic databases PubMed, IEEE and ACM Digital Library. The box below presents the search strings used for this systematic literature review. The search strings were developed based on the key research questions of the study (see **Error! Reference source not found.**) and following several discussions among the study team. The strings were tested to make sure valid and reliable outcomes were obtained. Studies published within the last 10 years (2015-2024) were considered, and all eligible studies had to be open source, the full text available, and be written in English.

***Box** 8. Research questions and query strings.*

Titles and abstracts of articles published between 2015 and 2024 were searched for the following terms in PubMed:

Research question #1: How can bioinformatics' innovation, including AI-based approaches, improve metagenomic analysis making it accessible for the wide medical community (accurate, fast, and easily interpretable analysis of metagenomic data)?

("artificial intelligence"[Title/Abstract] OR "AI"[Title/Abstract] OR "ML"[Title/Abstract] OR "bioinfo*"[Title/Abstract] OR "Machine Learning"[Title/Abstract] OR "Deep Learning"[Title/Abstract] OR "LLM"[Title/Abstract] OR "computational biology"[Title/Abstract] OR "health technolog*"[Title/Abstract] OR "big data*"[Title/Abstract]) AND ("health preparedness"[Title/Abstract] OR "health emergenc*"[Title/Abstract] OR "pathogen identif*"[Title/Abstract] OR "pathogens identif*"[Title/Abstract]OR "emerging pathogen*"[Title/Abstract] OR "emerging infectio*"[Title/Abstract] OR "community wastewater" [Title/Abstract] OR "clinical surveillance" [Title/Abstract] OR "pathogen surveillance" [Title/Abstract] OR "disease outbreak*" [Title/Abstract] OR "pandemic preparedness*" [Title/Abstract]) AND ("metagenomic*"[Title/Abstract] OR "NGS"[Title/Abstract] OR "genomic sequencing"[Title/Abstract] OR "deep sequencing"[Title/Abstract] OR "metabolomic*"[Title/Abstract])

Hits: 99

Research question #2: What are the bottlenecks of using metagenomics for emerging pathogen detection/health preparedness?

("limitation*"[Title/Abstract] OR "bottleneck*"[Title/Abstract] OR "barrier*"[Title/Abstract]) AND ("health preparedness"[Title/Abstract] OR "health emergenc*"[Title/Abstract] OR "pathogen identif*"[Title/Abstract] OR "pathogens identif*"[Title/Abstract]OR "emerging pathogen*"[Title/Abstract] OR "emerging infectio*"[Title/Abstract] OR "community wastewater" [Title/Abstract] OR "clinical surveillance" [Title/Abstract] OR "pathogen surveillance" [Title/Abstract]) AND ("metagenomic*"[Title/Abstract] OR "NGS"[Title/Abstract] OR "genomic sequencing"[Title/Abstract] OR "deep sequencing"[Title/Abstract] OR "metabolomic*"[Title/Abstract])

Hits: 32

Research question #3: How can AI-based tools reduce the storage and processing requirements of systematic metagenomic surveillance?

(“computational resources”[Title/Abstract] OR “data storage”[Title/Abstract] OR “transfer requirement*”[Title/Abstract]) AND ("artificial intelligence"[Title/Abstract] OR "AI"[Title/Abstract] OR "ML"[Title/Abstract] OR "bioinfo*"[Title/Abstract] OR "Machine Learning"[Title/Abstract] OR "Deep Learning"[Title/Abstract] OR "LLM"[Title/Abstract] OR "computational biology"[Title/Abstract] OR “health technolog*”[Title/Abstract] OR “big data*”[Title/Abstract]) AND ("health preparedness"[Title/Abstract] OR "health emergenc*"[Title/Abstract] OR "pathogen identif*"[Title/Abstract] OR "pathogens identif*"[Title/Abstract]OR “emerging pathogen*"[Title/Abstract] OR “emerging infectio*"[Title/Abstract] OR “community wastewater” [Title/Abstract] OR “clinical surveillance” [Title/Abstract] OR “pathogen surveillance” [Title/Abstract] OR "metagenomic*"[Title/Abstract] OR “NGS"[Title/Abstract] OR “genomic sequencing"[Title/Abstract] OR “deep sequencing"[Title/Abstract] OR “metabolomic*”[Title/Abstract])

Hits: 67

Titles and abstracts of articles published between 2015 and 2024 were searched for the following terms in IEEE:

("artificial intelligence" OR "AI" OR "ML" OR "Machine Learning" OR "Deep Learning" OR "LLM" OR "computational biology") AND ("emergenc*" OR "pathogen*" OR "surveillance" OR "outbreak*" OR "pandemic*") AND ("metagenomic*" OR "genomic sequencing")

Hits: 17

Titles and abstracts of articles published between 2015 and 2024 were searched for the following terms in ACM:

("artificial intelligence" OR "AI" OR "ML" OR "Machine Learning" OR "Deep Learning" OR "LLM" OR "computational biology") AND (("health preparedness" OR "health emergenc*" OR "pathogen identif*" OR "pathogens identif*"OR "emerging pathogen*" OR "emerging infectio*" OR "community wastewater" OR "clinical surveillance" OR "pathogen surveillance" OR "disease outbreak*" OR "pandemic preparedness*") OR ("metagenomic*" OR "NGS" OR "genomic sequencing" OR "deep sequencing" OR "metabolomic*"))

Hits: 47

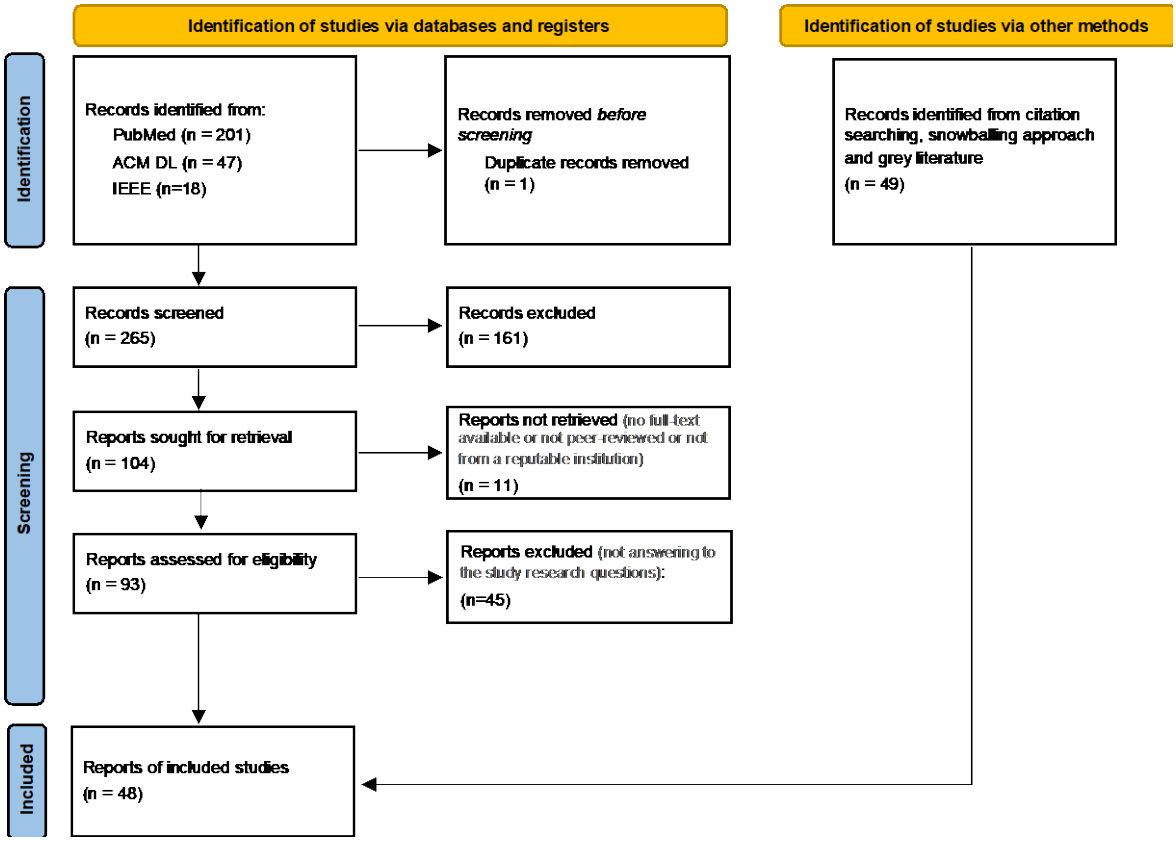
Search hits were downloaded, resulting in one file prepared for the screening of the search hits (n=265). Citations were screened by title and abstract to see if they explored 1) AI-based tools used for metagenomic analysis, 2) the bottlenecks of metagenomic analysis, or 3) AI-based solutions for metagenomic data storage and transferability. Based on these criteria, 104 publications were selected for further full-text review.

A quick full-text review was conducted before potentially eligible reports were further assessed. Studies were excluded if they were: 1) not peer-reviewed, 2) outside of the research scope, and 3) non-relevant to the research questions. A total of 48 studies were selected for data extraction. Reasons for exclusion for the other studies were recorded. The study team manually added 49 key references identified outside of the database search, which emerged from checking the references of relevant publications and were considered key (snowballing approach). A single reviewer extracted data from each included report. A second reviewer checked extractions and appraisals for accuracy and completeness. A narrative synthesis was performed.

The total number of records after each screening round was documented using the PRISMA flow diagram template (see Figure below). The review of the literature assisted the study team in identifying some authors that explored the contribution of AI-based tools to enhance

metagenomic analysis, analysed the bottlenecks in current metagenomics, and presented different options for innovations in the field of AI and metagenomics.

Figure 3. *PRISMA flow diagram summarising the records of the literature review.*



Source*: authors' elaboration based on Page et al. (2021)*

Consultation activities

The consultation activities were carried out for two months and aimed to achieve three primary objectives:

1. **Validate and bridge research gaps:** To corroborate and fill the research gaps identified in existing literature by integrating insights from experts across academia, public health laboratories, industry, and government.
2. **Identify and sort bottlenecks:** To pinpoint and rank the most critical bottlenecks and obstacles hindering progress in the field of metagenomics.
3. **Validate AI solutions:** To assess the validity and feasibility of AI solutions identified in the literature for metagenomic sequencing from both technical and clinical practice perspectives.

To meet these objectives, the consultations included in-depth one-on-one interviews (60 minutes long) and a structured survey. Prior to the consultations, four distinct interview guidelines were developed, each tailored to a specific stakeholder category (academia, public health laboratory, industry, and government). These guidelines were reviewed and validated by HERA, ensuring comprehensive coverage of key topics while allowing flexibility for individual expertise.

Each guideline included the following sections:

1. Introduction and consent for recording and minute-taking.

2. Background questions on the interviewee's profile and experience in the field.
3. Questions regarding the use of AI tools in the three main stages of the metagenomic workflow.
4. Questions exploring research bottlenecks and the maturity of existing tools.
5. Open-ended questions for additional insights, uncovered areas, or recommendations for further consultation.

A comprehensive stakeholder mapping exercise was conducted at the outset of the study (and lasted six months), identifying key experts across the four stakeholder categories. This mapping was validated by HERA and informed the subsequent engagement activities, which included targeted emails to request participation in interviews and the survey.

The consultation strategy envisioned a target number of participants for each category:

- Academia (N = 5)
- Public Health Laboratory (N = 2-5)
- Industry (N = 2-5)
- Government (N = 2-5)

During the two-month consultation period, other stakeholders from those originally selected were included in the engagement activities after a thorough review of their expertise. The table below summarises the number of stakeholders contacted for each consultation activity and the rate of positive answers.

Table 6. *Consultation activities figures.*

	Survey (N)	Interview (N)
Academia	85	22
Public health laboratories	9	7
Industry	16	5
Government	5	5

Ultimately, **two** stakeholders from academia, **two** from public health laboratories, **three** from industry, and **two** from government were consulted, providing invaluable feedback and input.

Interviews followed a semi-structured format, allowing for adaptation based on the interviewee's specific expertise. Preliminary materials, including tables of identified AI tools and research bottlenecks, were shared with interviewees a week in advance. Following each interview, detailed minutes were shared for review and approval before incorporating the feedback into the study.

The survey, administered through the EUSurvey tool, was distributed to all mapped experts, cross-checking that they were not also engaged in topic A2. Its structure mirrored the interview guide, with sections on general questions, AI tools in the metagenomic workflow, and research bottlenecks.

The valuable insights gathered from the consultation activities are presented at the end of each relevant results section (Box 2-6). These findings collectively contributed to the final version of the research bottleneck table in section 2.5 and were also reflected in the box in subsection 2.3. “Main outcomes resulting from consultation activities”, offering a comprehensive overview of the challenges and opportunities identified by stakeholders in the field of metagenomics.

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